SCHWARZ RESEARCH INTELLIGENCE

User Documentation & Operational Manual

Comprehensive Administrator & AI Agent Guide for Automated Systematic Literature Reviews, Dynamic Parameter Extractions & Technology Scouting

PRIMARY ENDPOINTS
https://research-scanner.schwarz / STACKIT Cloud

DEPLOYMENT ARCHITECTURE
Django Admin + FastMCP SSE + Next.js

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DOCUMENTATION MODULES
Guides 01–07 (Admin & MCP Dual Tracks)

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🚀 Schwarz Research Intelligence (SRI)

User Guide & Operational Manual

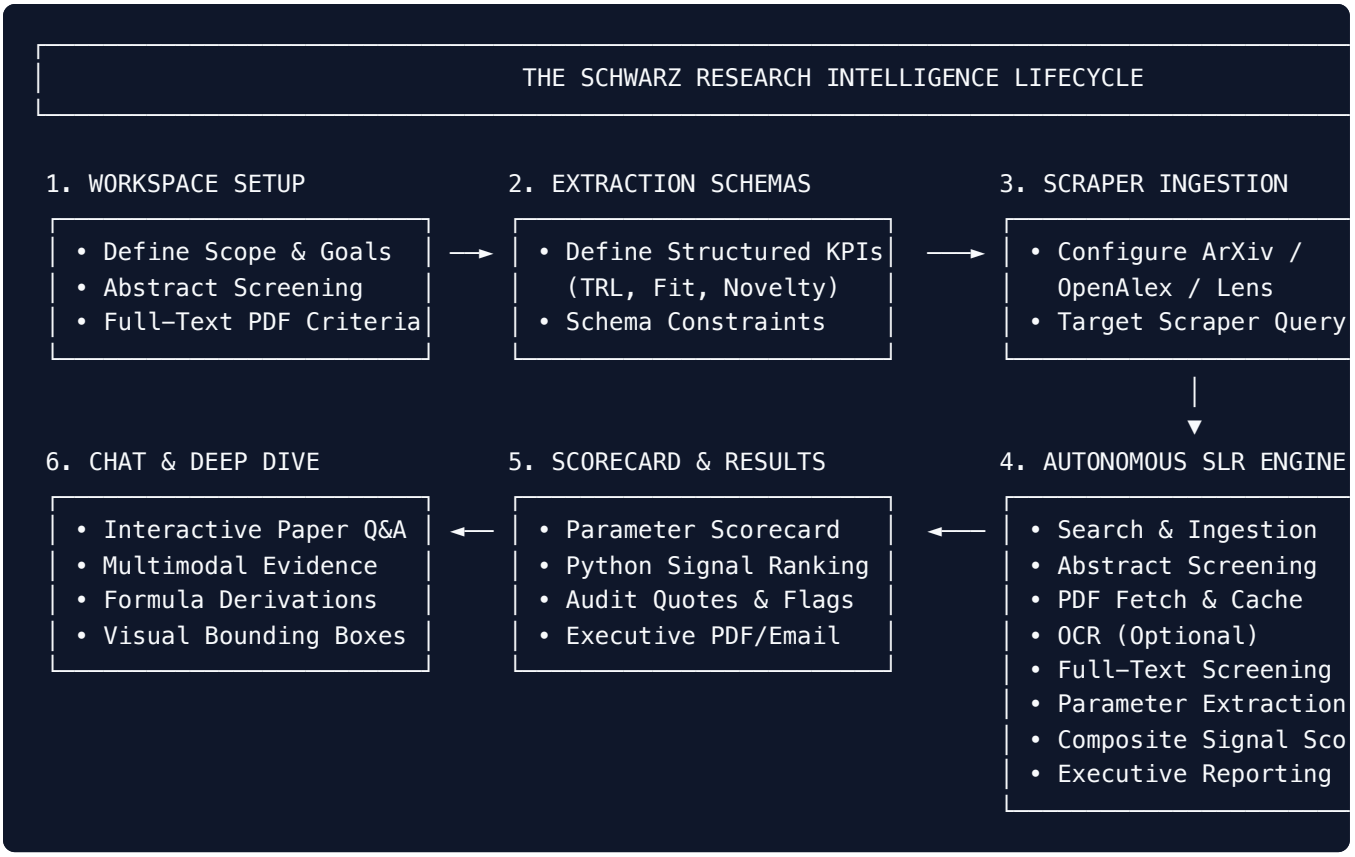
Welcome to **Schwarz Research Intelligence (SRI)** — an enterprise-grade AI intelligence platform engineered for automated Systematic Literature Reviews (SLR), deep-tech scouting, competitive intelligence, and high-precision parameter extraction.

🌐 Platform Access & Personas

Target Persona	Service & Access Channel	Endpoint URL	Description
👨‍💼 **Manager / Researcher**	**SRI Next.js Web App**	https://research-scanner.schwarz/ *(or Cloud Foundry mirror)*	Primary dashboard, research matrix, interactive scorecards, and Paper Chat.
🛡️ **Administrator / Lead**	**Django Administration Portal**	https://research-scanner.schwarz/admin/ *(or Cloud Foundry mirror)*	Full management panel for Workspaces, RBAC, Quotas, Audit Logs, and Storage Browser.
⚡ **Power User / AI Agent**	**FastMCP SSE API (Claude/Cursor)**	`https://research-scanner.schwarz/mcp/sse` *(or Cloud Foundry mirror)*	MCP endpoint for Claude Code, Cursor, Antigravity IDE, and autonomous agents.

🧭 End-to-End System Workflow

SRI automates the entire lifecycle from academic ingestion to custom scorecards and deep multi-turn paper interrogations:



📚 User Documentation Modules

Choose your preferred way of working — whether you prefer the visual **Django Admin Portal** or conversational **Claude Code / MCP AI Agents**:

1. ⚙️ Setup & Configuration

- **01. Pipeline Setup via Django Admin**
  - Step-by-step visual configuration: Creating Workspaces, adding team members and email subscribers, configuring dual-stage screening criteria, setting up data scrapers, defining dynamic extraction schemas, and writing Python AST scoring formulas.
- **02. Pipeline Setup via MCP & Claude Code**
  - Natural language and programmatic setup: Configuring your entire research review workspace in seconds through Claude Code or any MCP-compatible AI client.

2. ⚡ Launching & Monitoring Runs

- **03. Running Research Pipelines via Django Admin**
  - Triggering Pipeline Runs, monitoring real-time stage progress, inspecting the unified "Papers (In ➔ Out)" funnel, managing stopped stages, and executing bulk state machine actions.
- **04. Running Research Pipelines via MCP & Claude Code**
  - Executing reviews autonomously with `start_research_agent`, monitoring live ASCII progress bars via `check_run_status`, pausing, resuming, and aborting runs on demand.

3. 📊 Results & Technology Scorecards

- **05. Viewing Results & Scorecards via Django Admin**
  - Navigating screened papers, exploring LLM decision rationales, reviewing extracted KPI values with verbatim quotes and confidence scores, inspecting storage artifacts, and sending email digests.
- **06. Viewing Results & Scorecards via MCP & Claude Code**
  - Querying full aggregated technology matrices with `get_paper_extractions_matrix`, listing top-scoring research papers, and generating executive tabular comparisons.

4. 💬 Interactive Paper Chat & Deep Dive

- **07. Interrogating & Chatting with Papers via MCP**
  - Interacting with ingested academic papers using `ask_paper` powered by Google Gemini 3.7 Flash: deep mathematical derivations, experimental methodology validation, visual figure verification, and hallucination-guarded answers.

🎯 Key Benefits for Enterprise Research Teams

1. **Massive Time Savings**: Accelerate Systematic Literature Reviews from months of manual reading to under 15 minutes of autonomous processing.
2. **Zero Hallucinations**: Every extracted data point is tied to direct verbatim evidence quotes and visual coordinates in the original PDF.
3. **Custom Strategic Scoring**: Write transparent Python mathematical formulas to rank papers according to your business's proprietary TRL, strategic fit, and novelty thresholds.
4. **Multimodal Native Processing**: Directly analyzes tables, figures, charts, and equations using cutting-edge Google Gemini 3.7 Flash multimodal models.
5. **Collaborative Multi-Tenancy**: Granular role-based access (Admin, Manager, Researcher) with automatic email notification digests for team stakeholders.

🛠️ Guide 01: Setting Up a Research Pipeline via Django Admin

This guide provides a comprehensive walkthrough for researchers, project leads, and innovation managers setting up a Systematic Literature Review (SLR) and competitive technology scouting project using the **Schwarz Research Intelligence (SRI) Admin Portal**.

🌐 Quick Access

- **Web Application**: https://research-scanner.schwarz/ *(or Cloud Foundry mirror)*
- **Admin Portal URL**: https://research-scanner.schwarz/admin/ *(or Cloud Foundry mirror)*
- **Credentials**: Use your assigned Enterprise Single Sign-On (SSO) or administrative credentials (`admin` / `schwarz123!`).

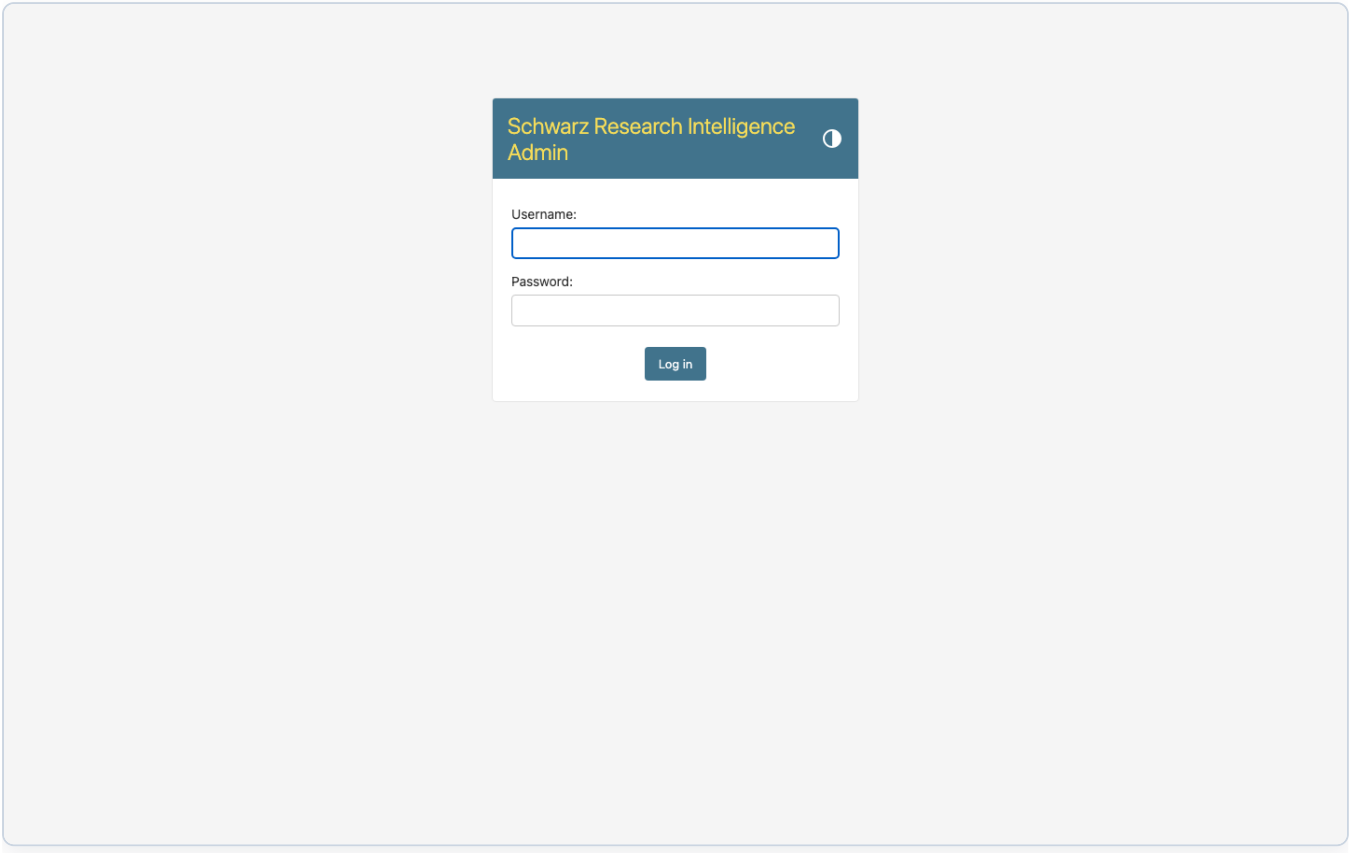
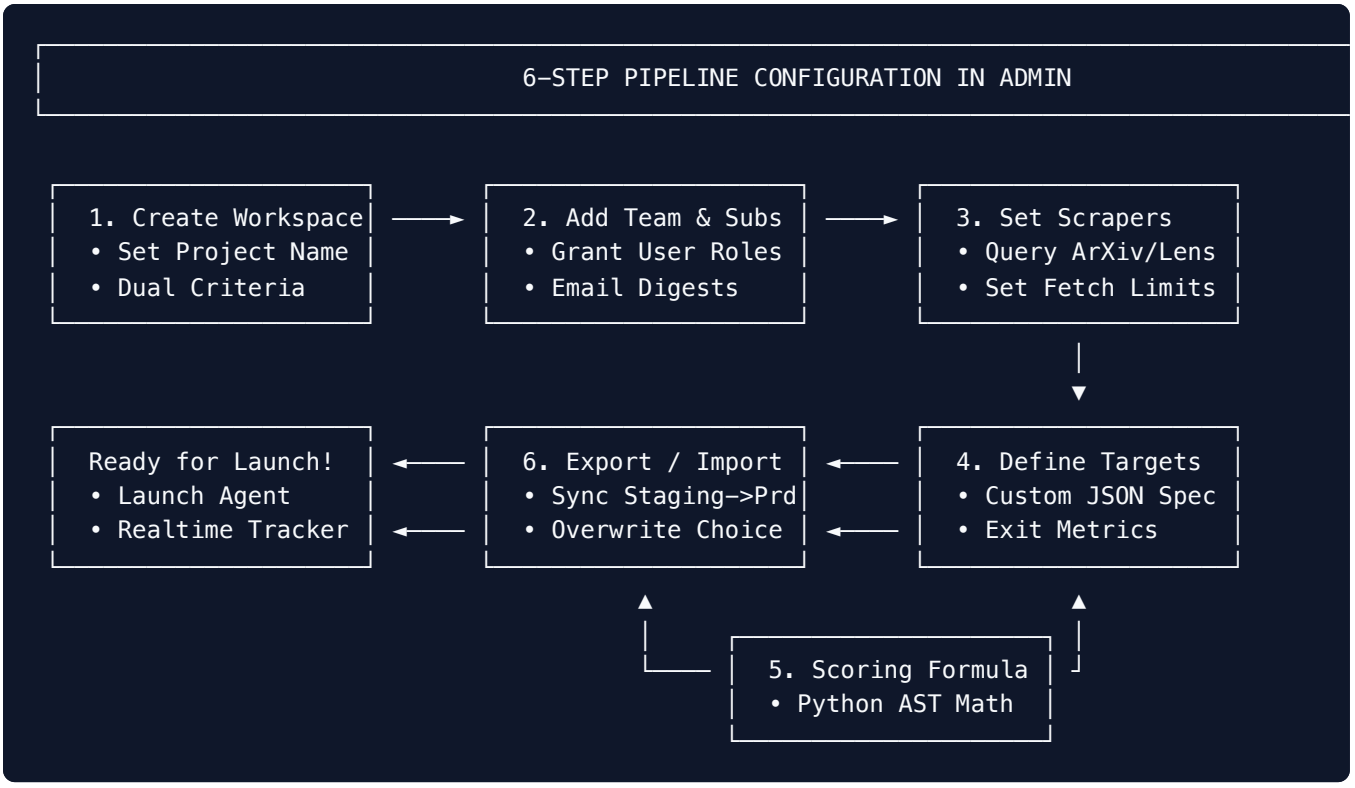


Figure 1: SRI Enterprise Single Sign-On and Administration Login Portal.

🗺️ Setup Workflow Overview

Setting up a complete intelligence pipeline involves 6 straightforward steps:



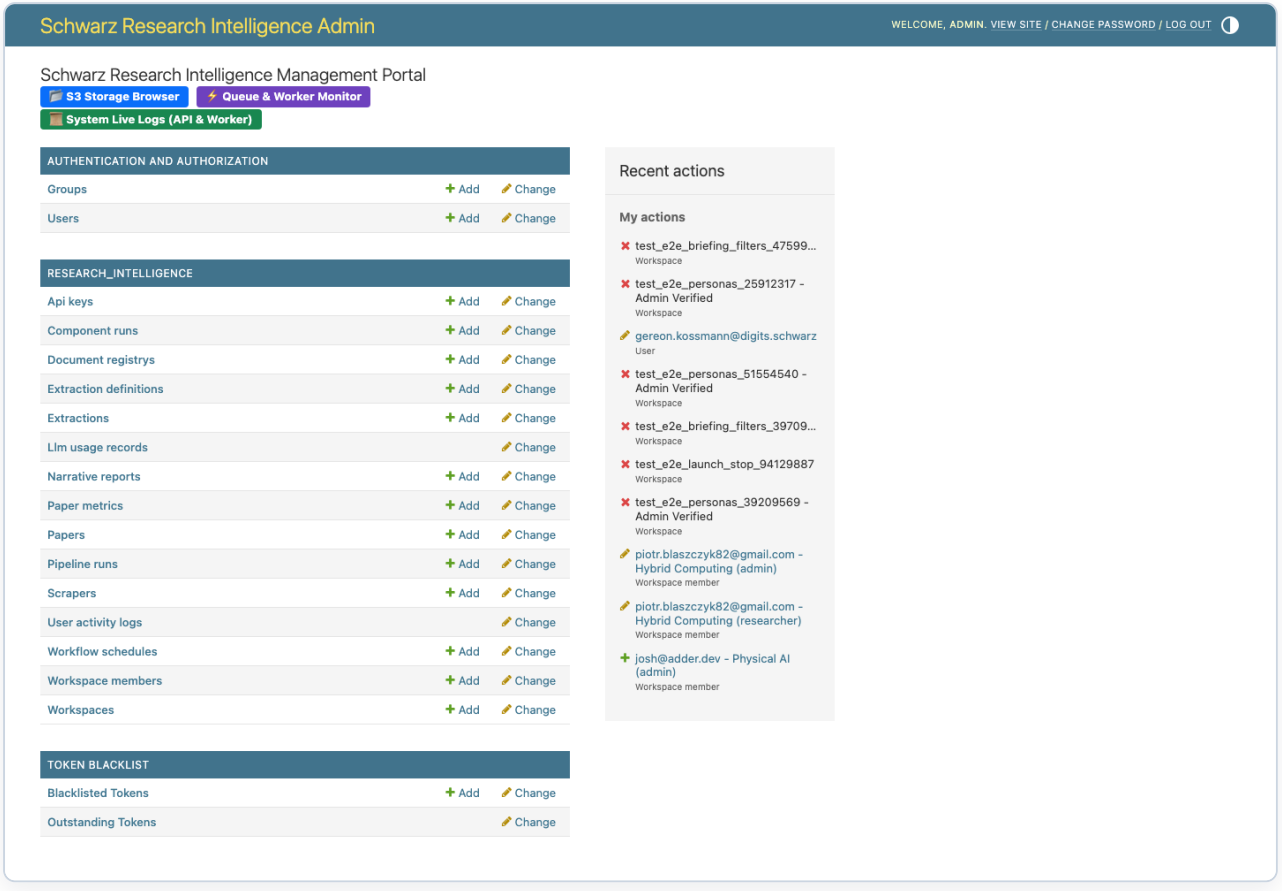


Figure 2: Production Django Admin Dashboard highlighting Workspaces, Scrapers, Extraction Definitions, and Pipeline Runs.

📋 Step 1: Create a Research Workspace

A **Workspace** is an isolated research initiative with dedicated inclusion/exclusion criteria, extraction rules, and signal scoring logic.

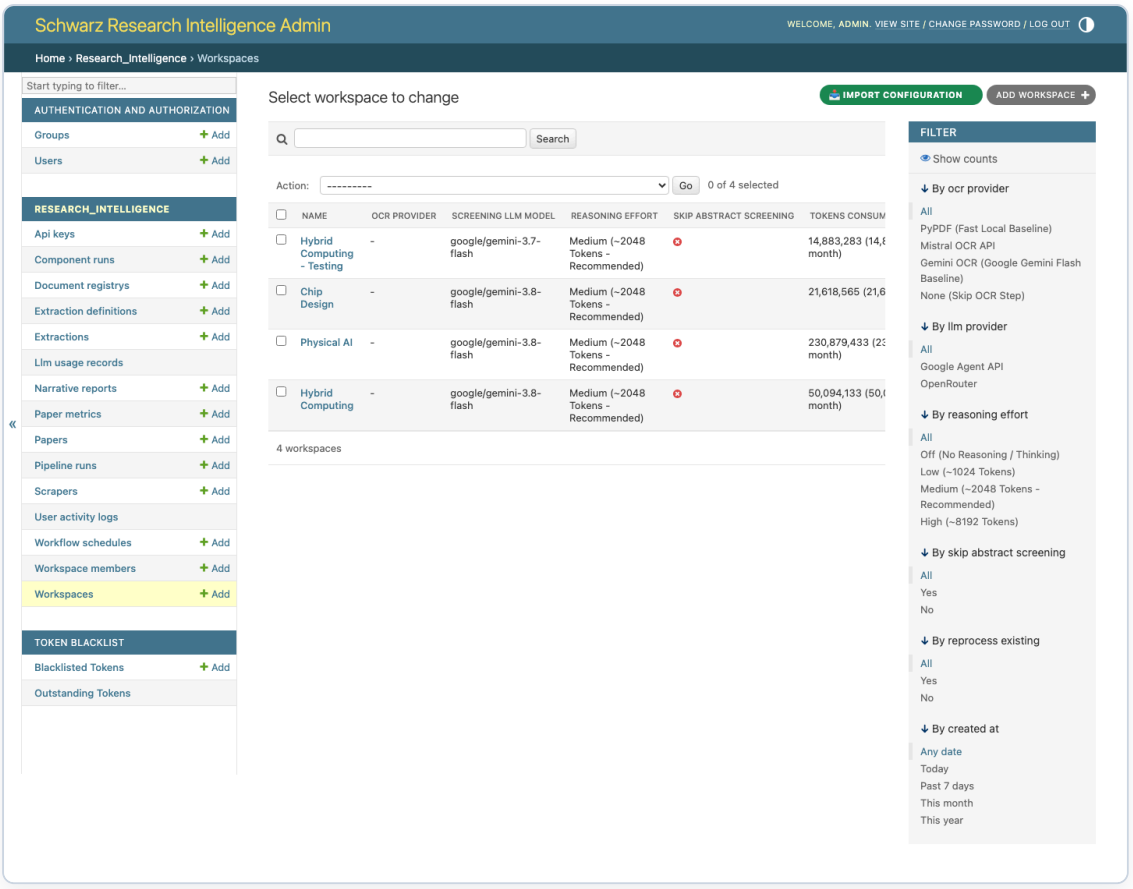


Figure 3: Active Workspaces list in SRI Admin.

1. Log into the SRI Admin Portal.
2. Under the **Research Intelligence** section, click **Workspaces** ➡️ **+ Add Workspace** (or edit an existing one).
3. Fill in the core project metadata:
   - **Name**: `Quantum Computing in Logistics & Supply Chain 2026`
   - **Reasoning Effort**: `medium` *(Chain of Thought thinking token budget for Gemini 3.7 Flash)*.

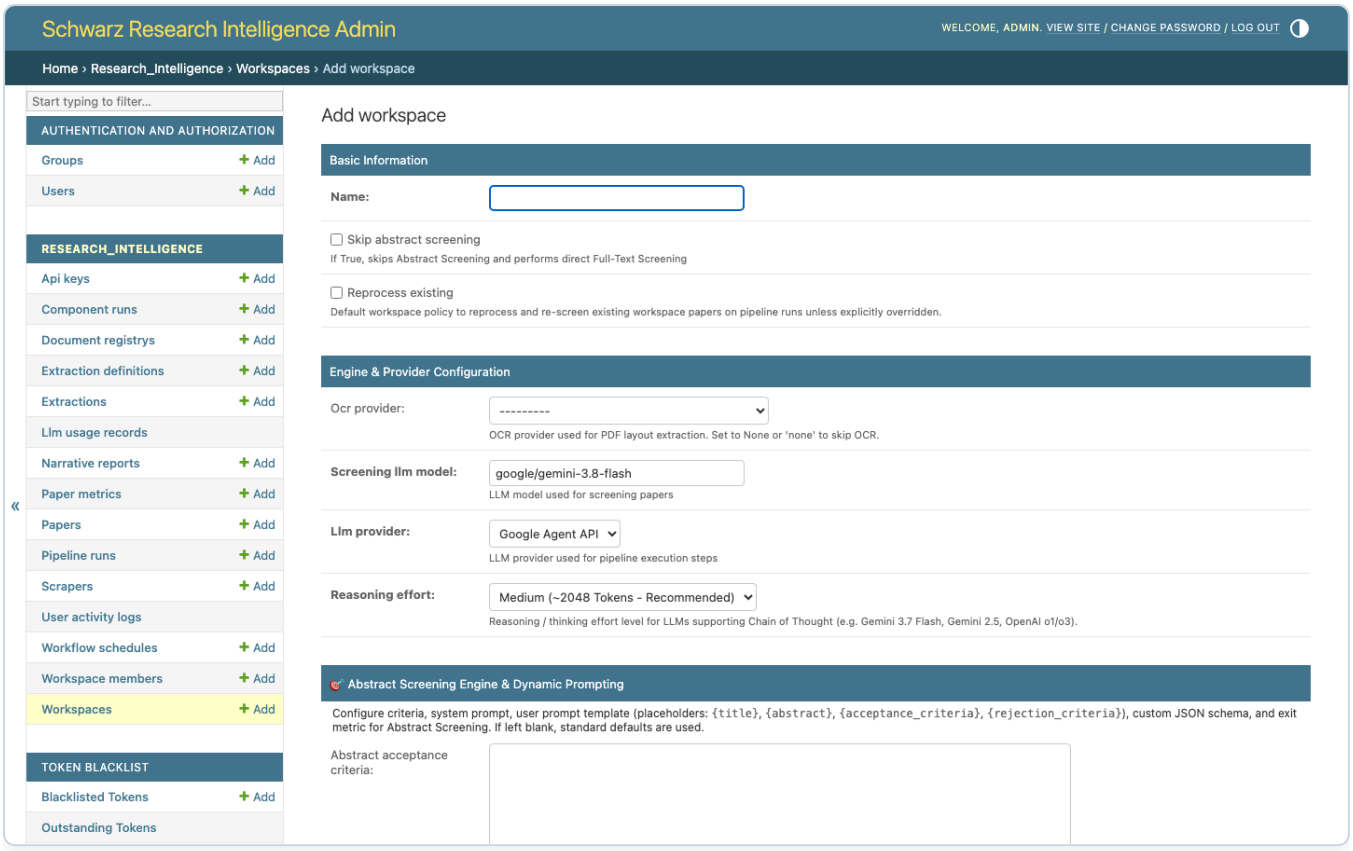


Figure 4: Workspace creation form defining project scope and reasoning effort.



Figure 5: Workspace configuration page with dual-stage screening criteria and Python signal score formula.

🎯 Configuring Dual-Stage Screening Criteria

SRI uses a two-stage screening funnel to save time and compute costs.

A. Stage 1: Abstract Screening Criteria (Fast Filter)

Evaluated strictly against paper titles and abstracts before downloading PDFs.

- **Abstract Acceptance Criteria**:

```
Accept papers presenting quantum algorithms (VQE, QAOA, quantum annealing, or fault-tole
```

- **Abstract Rejection Criteria**:

```
Reject pure hardware physics papers without algorithmic application, purely classical he
```

B. Stage 2: Full-Text Screening Criteria (Deep Multimodal Verification)

Evaluated directly against the raw full-length PDF document using multimodal Gemini 3.7 Flash.

- **Full-Text Acceptance Criteria**:

```
Accept only if the full paper provides explicit mathematical formulations of the problem
```

- **Full-Text Rejection Criteria**:

```
Reject if the paper only mentions logistics as an abstract future possibility, lacks exp
```

C. Engine, Provider & Processing Policies

- **Screening LLM Model**: `google/gemini-3.8-flash` *(Recommended default for ultra-fast, cost-effective multimodal analysis)*.
- **OCR Provider**: `None` *(Leave as `None` to pass native PDF binaries directly into Gemini's native multimodal visual reasoning engine; select `pypdf`, `mistral`, or `gemini` only if raw Markdown text layout is required)*.
- **Reprocess Existing**: `False` *(Default workspace policy. If checked/True, pipeline runs default to re-evaluating and re-screening existing workspace papers even if no new papers are found)*.
- **Skip Abstract Screening**: `False` *(If checked/True, bypasses Abstract Screening and routes all ingested candidate papers directly to deep Full-Text Screening)*.
- **Cost Limit (USD)**: `50.00` *(Safety ceiling per run)*.

Click **Save and continue editing**.

👥 Step 2: Add Team Members, Email Subscribers & Workflow Schedules

Collaborate securely with team members, configure automated briefing delivery, and establish automated recurring review schedules.

Adding Team Members

1. Scroll down to the **Workspace Members** inline table (or navigate to **Workspace Members** in the sidebar).
2. Click **+ Add another Workspace Member**:
   - **User**: Select the team member's username or email.
   - **Role**:
     - `Admin`: Full rights to modify criteria, delete runs, and manage members.
     - `Manager`: Can launch runs, edit scrapers, and update extraction definitions.
     - `Researcher`: Read-only access to scorecards, chat explorer, and downloads.

Adding Notification Subscribers

Keep executive stakeholders informed automatically upon pipeline completion:

- In the **Workspace Subscribers** inline table, add the email addresses of stakeholders and choose their **Subscription Type**:
  - `Result Report (Scorecard & Findings)`: Receives parameter scorecards and top papers.
  - `Pipeline Errors (Failure Alerts)`: Receives alerts if an issue occurs.
  - `Pipeline Status (Execution Digest)`: Receives step progress reports.
  - `All Notifications (Full Broadcast)`: Full broadcast of all run events.

Configuring Automated Workflow Schedules

Establish autonomous recurring runs:

- In the **Workflow Schedules** inline table, click **+ Add another Workflow Schedule**:
  - **Name**: e.g., `Daily 3pm Ingestion` or `Weekly Deep Tech Review`.
  - **Frequency**: `Daily`, `Weekdays (Mon–Fri)`, `Weekly`, `Monthly`, `Custom Cron`, or `One-Off`.
  - **Reprocess Existing**: Optionally enable reprocessing for this scheduled job.
  - **Limit & Date Range**: Restrict candidate paper count and publication date window per run.

🔍 Step 3: Configure Academic Scrapers

Scrapers define what academic databases to search and what query strings to execute during Step 1 of the pipeline.



Figure 6: Active Scrapers configured across ArXiv, OpenAlex, and Lens.

1. In the Admin sidebar, click **Scrapers** ➡️ **+ Add Scraper** (or use the inline section on the Workspace page).

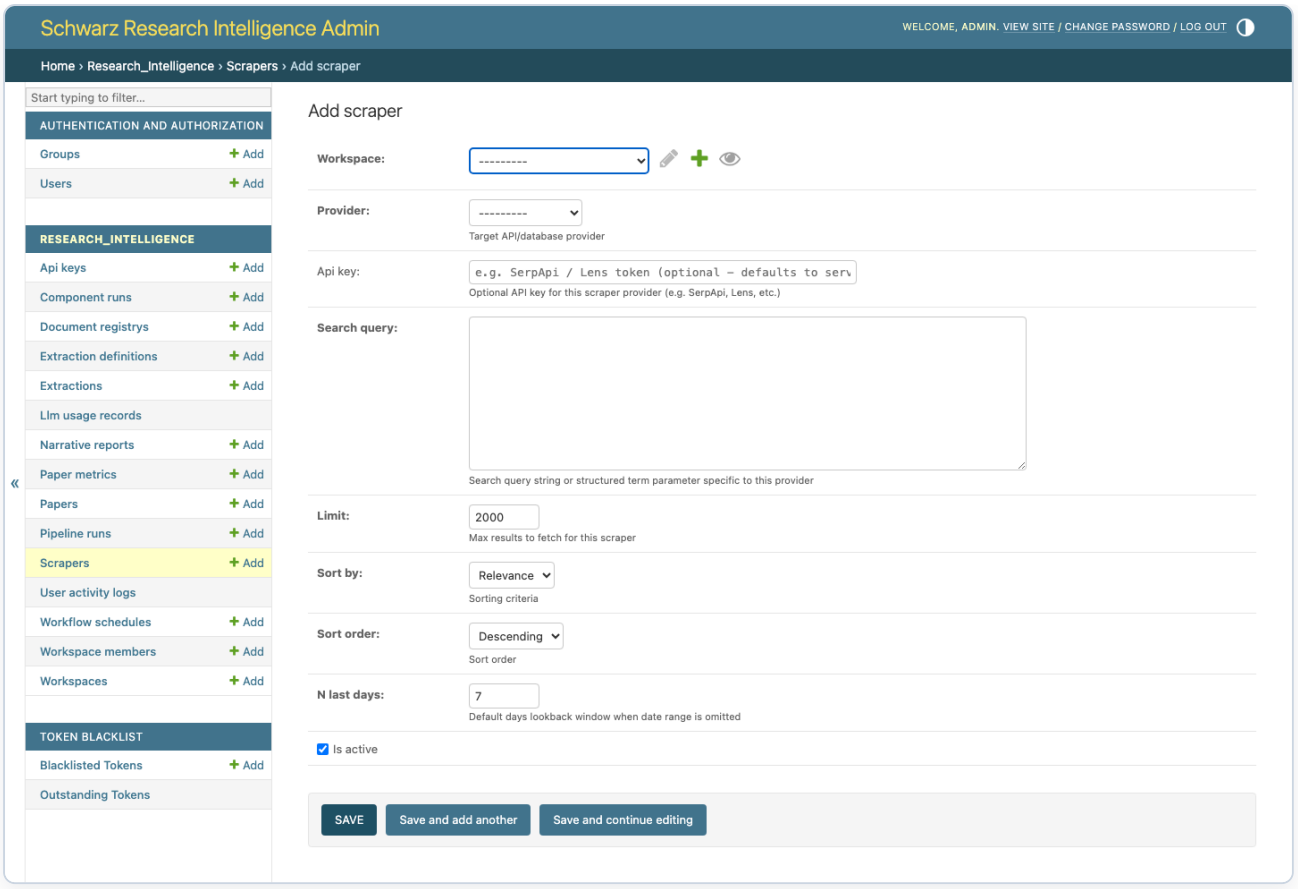


Figure 7: Scraper configuration form with query, candidate limits, lookback days, and provider selection.

2. Set the configuration:
   - **Workspace**: Select your workspace.
   - **Provider**:
     - `arxiv` — ArXiv e-Prints (Physics, CS, Math, Quant-Bio).
     - `openalex` — Global bibliographic catalog across 250M+ scientific works.
     - `lens` — Scholarly literature and global patent databases.
   - **Search Query**:
     - *For ArXiv*: `cat:quant-ph AND (all:"supply chain" OR all:"vehicle routing" OR all:"logistics")`
     - *For OpenAlex*: `quantum computing logistics vehicle routing optimization`
   - **Limit**: `100` *(Target number of candidates to fetch)*.
   - **N Last Days**: `7` *(Default lookback window in days when running search without explicit start/end dates. Defaults to 7)*.
   - **Is Active**: `Checked` (True).
3. Click **Save**.

📊 Step 4: Define Dynamic Parameter Extraction Targets

Extraction definitions tell the AI what specific parameters, KPI values, and technology metrics to extract from full-text papers that pass dual-stage screening.

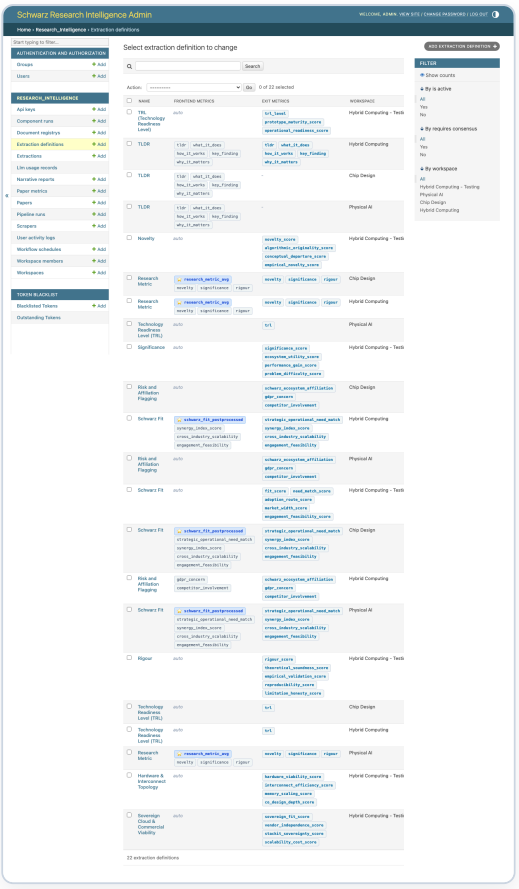


Figure 8: Extraction Definitions configured with exit metrics and custom LLM schemas.

1. Navigate to **Extraction Definitions** ➡️ **+ Add Extraction Definition** (or edit an existing one).

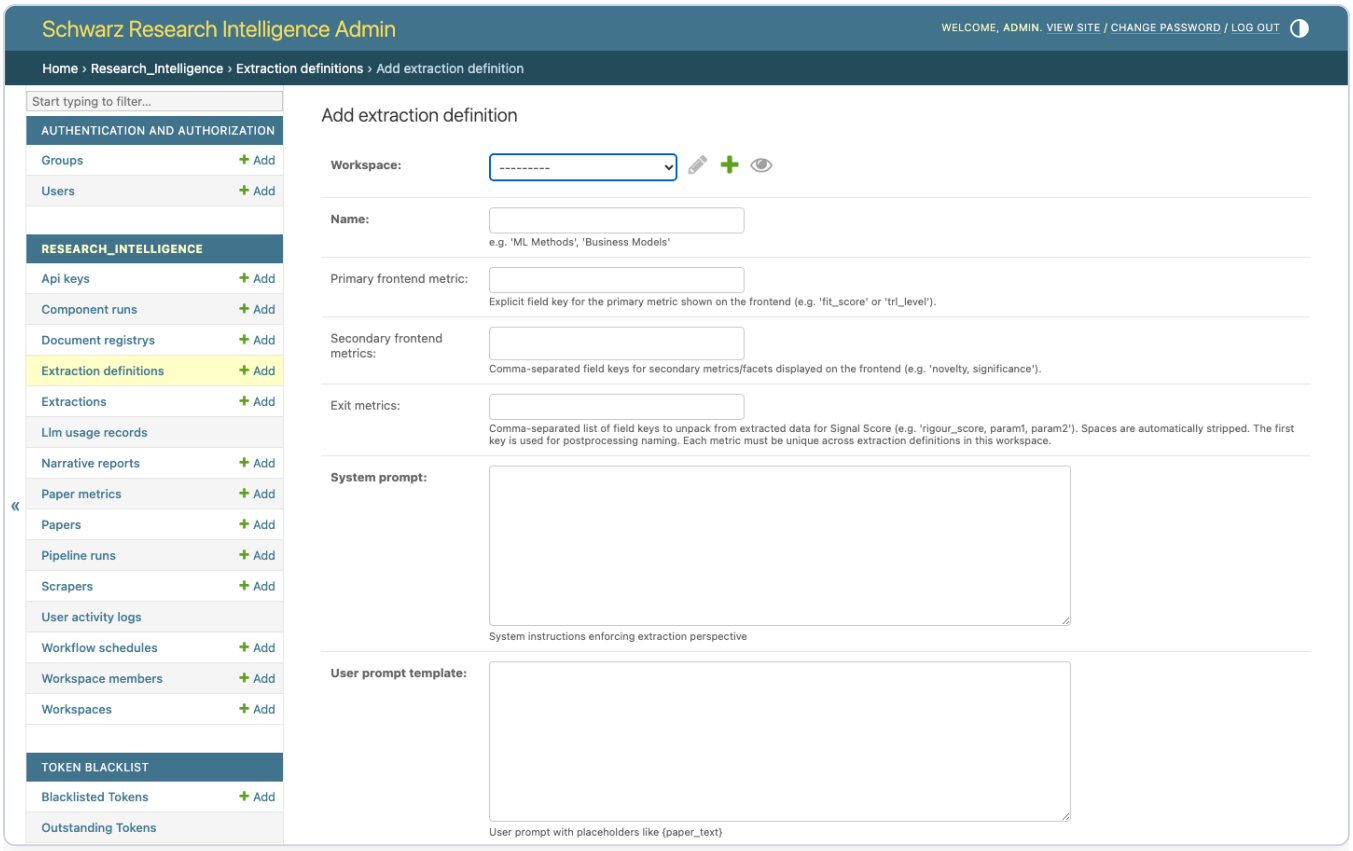


Figure 9: New Extraction Definition creation modal.

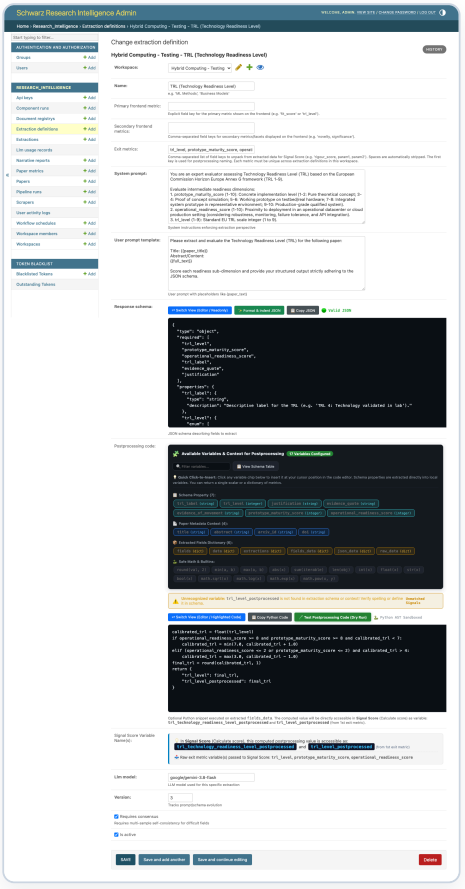


Figure 10: JSON Schema and System Prompt editor for structured parameter extraction.

2. Example Target 1: **Schwarz Strategic Fit & Application Scope**

- **Name**: `Schwarz Strategic Fit`
- **Exit Metric**: `fit_score` *(Unique key for matrix display and scoring)*
- **System Prompt**:

```
You are an expert technology scout for the Schwarz Group. Evaluate how directly appli
```

- **User Prompt Template**:

```
Evaluate the following paper for enterprise logistics application:
Title: {{paper_title}}
Abstract: {{abstract}}
Full Text: {{full_text}}
```

- **Response Schema (JSON Schema)**:

```
{
  "type": "object",
  "properties": {
    "fit_score": { "type": "number", "minimum": 1, "maximum": 10 },
    "primary_use_case": { "type": "string" },
    "algorithmic_family": { "type": "string", "enum": ["QAOA", "VQE", "Quantum Anneal
    "scalability_assessment": { "type": "string" },
    "verbatim_quote": { "type": "string" }
  },
  "required": ["fit_score", "primary_use_case", "algorithmic_family", "verbatim_quote
}
```

3. Example Target 2: **Technology Readiness Level (TRL)**

- **Name**: `TRL Assessment`
- **Exit Metric**: `trl_level`
- **Response Schema**:

```
{
  "type": "object",
  "properties": {
    "trl_level": { "type": "number", "minimum": 1, "maximum": 9 },
    "experimental_validation": { "type": "string" },
    "hardware_platform_used": { "type": "string" },
    "qubit_count": { "type": "integer" }
  },
  "required": ["trl_level", "experimental_validation"]
}
```

🧱 Step 5: Configure Custom Python Signal Score Code

SRI enables you to write custom Python mathematical ranking formulas to prioritize papers automatically based on extracted parameters.

1. In your Workspace edit page, locate the **Signal Score Code** field.
2. Enter your custom mathematical AST formula. Extracted parameters are automatically exposed as sanitized `snake_case` variables:

```
# Weighted composite calculation combining Strategic Fit, TRL, and Novelty
# Minimum TRL gating: Projects below TRL 2 receive zero priority
if trl_level < 2:
    return 0.0

# Base calculation
score = (0.5 * fit_score) + (0.3 * novelty_score) + (0.2 * (trl_level * 1.1))

# High-qubit bonus multiplier
if qubit_count and qubit_count >= 50:
    score *= 1.15

return round(min(10.0, score), 2)
```

3. Click **Save**.

📦 Step 6: Export & Import Configurations (Staging ➔ Production Sync)

When you design, test, and fine-tune a workspace on **Staging**, you can seamlessly export and import the entire configuration to **Production**:

Exporting from Staging

1. On Staging Admin, go to **Workspaces**.
2. Check the box next to your workspace(s).
3. Under the **Action** dropdown, select 📦 **Export Workspace Configuration (JSON)** and click **Go**.
4. A complete `.json` configuration file is downloaded to your computer containing:
   - Dual-stage screening criteria & dynamic prompts
   - All Extraction Definitions and JSON schemas
   - Scrapers and search queries
   - Subscribers and email alerts
   - Custom Python Signal Score AST formulas

Importing to Production

1. On Production Admin, go to **Workspaces**.
2. Click the green 📩 **Import Configuration** button in the top right.

3. Upload the `.json` file (or paste JSON content directly).
4. Choose the overwrite mode:
   - **YES — Overwrite & Fine-Tune Existing Items** *(Recommended)*: Updates existing workspace rules, criteria, and extraction definitions in-place without losing paper history or run data.
   - **NO — Do Not Overwrite (Skip Existing Items)**: Only creates new workspaces and items.
5. Click 📩 **Import / Apply Configuration**.

🎉 Workspace Ready for Ingestion!

Your workspace is now fully configured. Next steps:

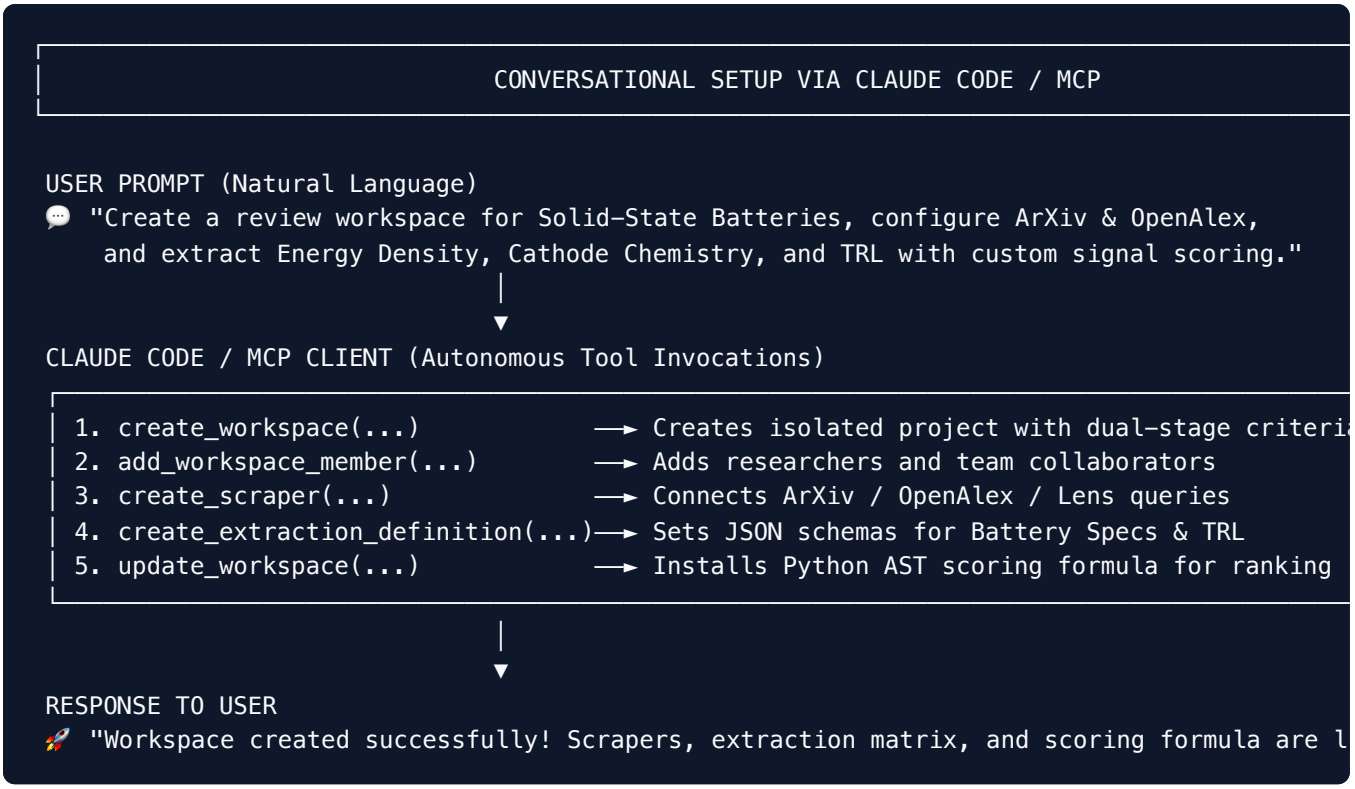
- To run your first autonomous review from the UI, see **Guide 03: Running Pipelines via Admin**.
- To launch and monitor reviews via Claude Code or MCP, see **Guide 04: Running Pipelines via MCP**.

🤖 Guide 02: Setting Up a Research Pipeline via MCP & Claude Code

With the **Schwarz Research Intelligence (SRI) Model Context Protocol (MCP)** server, you can configure your entire systematic review and competitive scouting pipeline conversationally in seconds using Claude Code, Cursor, or any MCP-enabled AI client.

🌐 MCP Connection & Architecture

- **Live MCP SSE Endpoint**: `https://research-scanner.schwarz/mcp/sse` *(or `https://sri-prod.apps.01.cf.eu01.onstackit.cloud/mcp/sse`)*
- **Web Application**: https://research-scanner.schwarz/ *(or Cloud Foundry mirror)*
- **Admin Portal**: https://research-scanner.schwarz/admin/ *(or Cloud Foundry mirror)*



💡 Quick Start: Single Conversational Prompt

You can set up an entire research initiative with a single natural language instruction in Claude Code:

> **User Prompt to Claude Code**: *"Please set up a new research workspace named 'Solid-State Battery Breakthroughs 2026'. Accept papers focusing on solid electrolyte materials, sulfide/oxide chemistries, and cell-level energy density > 400 Wh/kg. Reject pure liquid electrolyte or lead-acid chemistries. Configure scrapers for ArXiv and OpenAlex, create extraction definitions for 'Battery Specs' and 'TRL', and write a signal scoring formula prioritizing energy density and cycle life."*

Claude Code will automatically orchestrate the corresponding FastMCP tools step-by-step.

🛠️ Step-by-Step MCP Tool Execution

If you prefer to review or execute individual MCP tool calls, follow this sequence:

Step 1: Create the Isolated Workspace (`create_workspace`)

Creates the project with tailored dual-stage screening criteria and default reprocessing policies.

```
{
  "name": "Solid-State Battery Breakthroughs 2026",
  "abstract_acceptance_criteria": "Accept research papers evaluating all-solid-state lithiu
  "abstract_rejection_criteria": "Reject conventional liquid electrolyte lithium-ion papers
  "full_text_acceptance_criteria": "Accept if the full PDF reports verified experimental da
  "full_text_rejection_criteria": "Reject theoretical reviews without experimental or numer
  "screening_llm_model": "google/gemini-3.8-flash",
  "ocr_provider": null,
  "reprocess_existing": false
}
```

Tool Returns: `{"status": "success", "workspace_id": "7b8e1a90-3f2d-4b5c-8a1e-9d0c2e3f4a5b"}`

Step 2: Add Team Members (`add_workspace_member`)

Grant workspace permissions to your research team:

```
{
  "workspace_id": "7b8e1a90-3f2d-4b5c-8a1e-9d0c2e3f4a5b",
  "member_username_or_email": "dr.smith@schwarz.com",
  "role": "manager"
}
```

Step 3: Configure Academic Scrapers (`create_scraper`)

Set up multi-provider academic search queries to harvest relevant literature. Each scraper supports `n_last_days` (default `7`), defining the lookback window when runs are triggered without explicit start/end dates:

ArXiv Scraper

```
{
  "workspace_id": "7b8e1a90-3f2d-4b5c-8a1e-9d0c2e3f4a5b",
  "provider": "arxiv",
  "search_query": "cat:cond-mat.mtrl-sci AND (all:\"solid state battery\" OR all:\"solid el
  "limit": 50,
  "n_last_days": 7,
  "is_active": true
}
```

OpenAlex Scraper

```
{
  "workspace_id": "7b8e1a90-3f2d-4b5c-8a1e-9d0c2e3f4a5b",
  "provider": "openalex",
  "search_query": "all-solid-state lithium battery energy density ionic conductivity",
  "limit": 50,
  "n_last_days": 7,
  "is_active": true
}
```

Dry-Run Query Testing (`preview_search_query`)

Inspect live search results and tune queries with optional date filters before persisting a scraper:

```
{
  "provider": "arxiv",
  "search_query": "cat:cond-mat.mtrl-sci AND (all:\"solid state battery\")",
  "start_date": "2026-08-01",
  "end_date": "2026-08-31",
  "n_last_days": 14,
  "limit": 5
}
```

Step 4: Define Dynamic Extraction Schemas (`create_extraction_definition`)

Specify the exact structured JSON fields the AI must extract from screened full-text papers.

Extraction Target 1: Battery Performance Specifications (Multi-Metric & Postprocessing)

```
{
  "workspace_id": "7b8e1a90-3f2d-4b5c-8a1e-9d0c2e3f4a5b",
  "name": "Battery Specs",
  "exit_metrics": ["energy_density_wh_kg", "room_temp_conductivity_ms_cm", "cycle_retention_
  "postprocessing_code": "return (energy_density_wh_kg * 0.7) + (room_temp_conductivity_ms_
  "system_prompt": "You are an expert electrochemist and materials scientist. Extract quant
  "user_prompt_template": "Extract key electrochemical metrics from the paper:\nTitle: {{pa
  "response_schema": {
    "type": "object",
    "properties": {
      "electrolyte_type": { "type": "string", "enum": ["Sulfide", "Oxide/Garnet", "Polymer"
      "energy_density_wh_kg": { "type": "number", "description": "Gravimetric energy densit
      "room_temp_conductivity_ms_cm": { "type": "number", "description": "Ionic conductivit
      "cycle_retention_pct": { "type": "number", "description": "Capacity retention percent
      "cycles_tested": { "type": "integer", "description": "Number of discharge cycles test
      "evidence_quote": { "type": "string", "description": "Verbatim quote confirming the m
    },
    "required": ["electrolyte_type", "energy_density_wh_kg", "evidence_quote"]
  }
}
```

(Note: `exit_metrics` unpacks all 3 variables for Signal Score, and `postprocessing_code` creates `energy_density_wh_kg_postprocessed`).

Extraction Target 2: Technology Readiness Level (TRL)

```
{
  "workspace_id": "7b8e1a90-3f2d-4b5c-8a1e-9d0c2e3f4a5b",
  "name": "TRL Assessment",
  "exit_metrics": ["trl_level"],
  "system_prompt": "Assess the Technology Readiness Level (1-9) of the battery prototype.",
  "user_prompt_template": "Assess TRL for {{paper_title}}:\n{{full_text}}",
  "response_schema": {
    "type": "object",
    "properties": {
      "trl_level": { "type": "integer", "minimum": 1, "maximum": 9 },
      "manufacturing_readiness": { "type": "string" },
      "commercial_scalability": { "type": "string" }
    },
    "required": ["trl_level"]
  }
}
```

Step 5: Install Custom Python Signal Scoring Formula (`update_workspace`)

Define your custom mathematical ranking formula using the extracted metric variables (`energy_density_wh_kg`, `trl_level`, `cycle_retention_pct`):

```
{
  "workspace_id": "7b8e1a90-3f2d-4b5c-8a1e-9d0c2e3f4a5b",
  "signal_score_code": "if energy_density_wh_kg < 300:\n    return 0.0\nbase = (energy_dens
}
```

Step 6: Verify Configuration (`get_workspace`)

Inspect your complete live workspace configuration:


```
{
  "workspace_id": "7b8e1a90-3f2d-4b5c-8a1e-9d0c2e3f4a5b"
}
```

🚀 What's Next?

Your research intelligence workspace is fully configured. You can now:

- Launch your automated review run: **Guide 04: Running Pipelines via MCP**
- Inspect scorecard results: **Guide 06: Viewing Results via MCP**
- Ask questions directly to papers: **Guide 07: Chatting with Papers via MCP**

🚀 Guide 03: Running a Research Pipeline via Django Admin

This guide explains how to trigger, monitor, and manage end-to-end autonomous research runs using the consolidated **Pipeline Runs** interface in the **Schwarz Research Intelligence (SRI) Admin Portal**.

🌐 Quick Access

- **Admin Portal URL**: https://research-scanner.schwarz/admin/ *(or Cloud Foundry mirror)*
- **Pipeline Runs Direct URL**: https://research-scanner.schwarz/admin/research_intelligence/pipelinerun/

> **NOTE**
> **Unified Architecture**: Research Agent Processes have been fully consolidated into **Pipeline Runs**. All execution tracking, stage transitions, stopped-stage breakpoints, funnel statistics, and bulk resume controls are unified under a single model. Legacy URLs (`/admin/research_intelligence/researchagentprocess/`) automatically redirect to Pipeline Runs.

🔄 Autonomous Execution Architecture

When you trigger a run, the SRI Autonomous State Engine coordinates an 8-stage Directed Acyclic Graph (DAG) pipeline across Celery background workers:

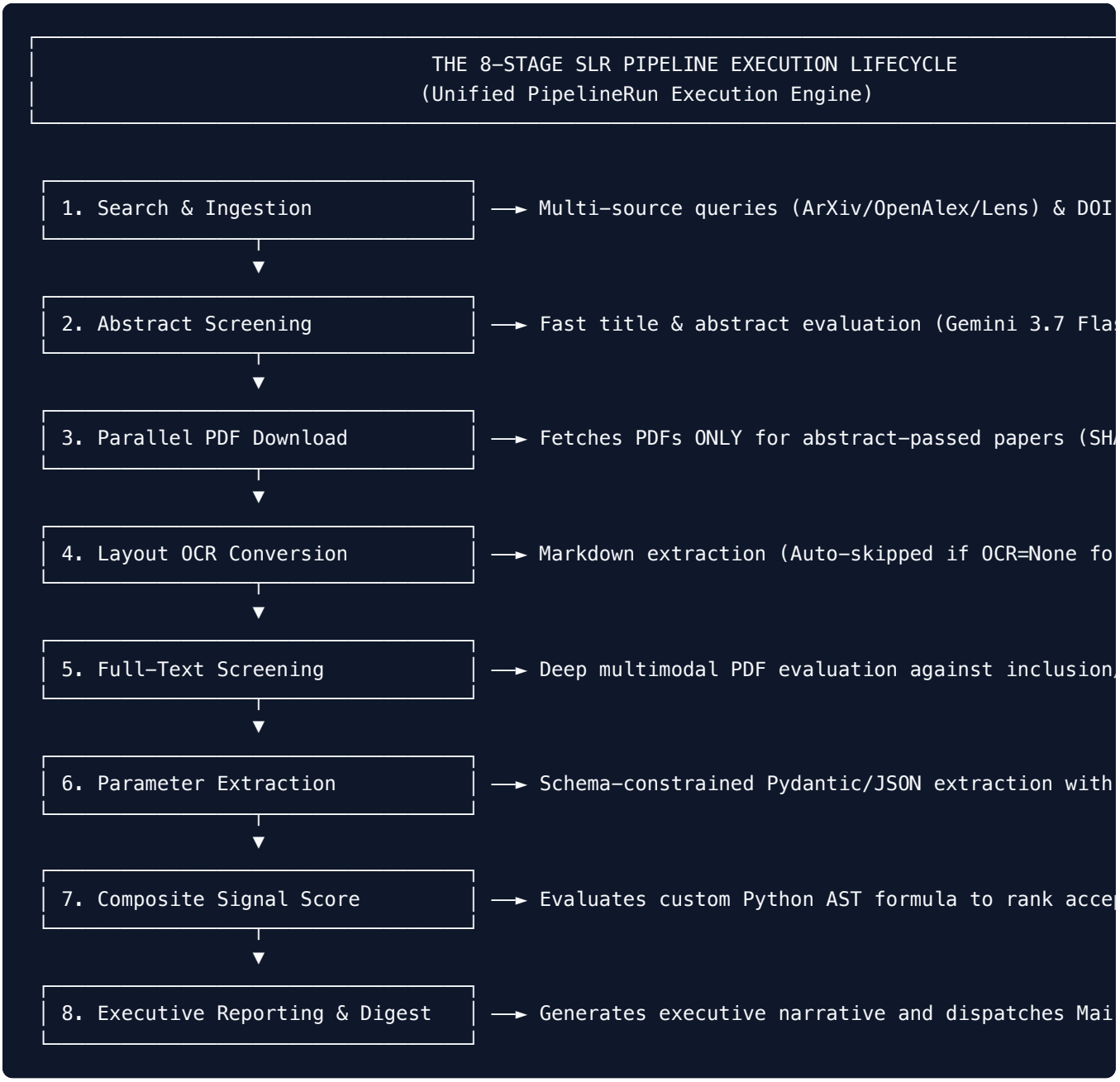

THE 8-STAGE SLR PIPELINE EXECUTION LIFECYCLE
(Unified PipelineRun Execution Engine)
1. Search & Ingestion
Multi-source queries (ArXiv/OpenAlex/Lens) & DOI
2. Abstract Screening
Fast title & abstract evaluation (Gemini 3.7 Fla
3. Parallel PDF Download
Fetches PDFs ONLY for abstract-passed papers (SH
4. Layout OCR Conversion
Markdown extraction (Auto-skipped if OCR=None fo
5. Full-Text Screening
Deep multimodal PDF evaluation against inclusion
6. Parameter Extraction
Schema-constrained Pydantic/JSON extraction with
7. Composite Signal Score
Evaluates custom Python AST formula to rank acce
8. Executive Reporting & Digest
Generates executive narrative and dispatches Mai



Figure 1: Consolidated Pipeline Runs changelist featuring the unified "Papers (In ➔ Out)" funnel column, stopped stage badges, and duration metrics.

⚡ Step-by-Step: Launching a Pipeline Run

1. Log into the Admin Portal.
2. In the sidebar under **Research Intelligence**, click **Pipeline runs** ➡️ **+ Add Pipeline run** (or inspect an existing run).
3. Configure run parameters:
   - **Workspace**: Select your target workspace (e.g. `Hybrid Computing` or `Quantum Computing in Logistics 2026`).
   - **Run Name**: Optional human-readable name (e.g. `Q1-2026 Comprehensive Review` or `Autonomous Research Agent – 2026-09-03 10:40`).
   - **Reprocess Existing**: `Unchecked` for incremental scouting; check this to re-evaluate and re-extract parameters from previously fetched papers with updated criteria.
   - **Stopped at Stage**: Optional execution breakpoint. Set to `abstract_screening` to pause execution after initial screening for human review, or leave empty for full end-to-end execution through executive reporting.
   - **Configuration Override**: Optional JSON payload to dynamically customize execution parameters, including scraper date range filtering:

```
{
  "start_date": "2026-08-01",
  "end_date": "2026-08-31",
  "n_last_days": 14,
  "limit": 50
}
```

- **`start_date`**: ISO date string (`YYYY-MM-DD`). Ingests literature submitted/published on or after this date.
- **`end_date`**: ISO date string (`YYYY-MM-DD`). Ingests literature submitted/published on or before this date. Defaults to `now()` if omitted when `start_date` is given.
- **`n_last_days`**: Number of historical days to crawl (`now() - N days` to `now()`). Defaults to **7 days** if no dates are specified.

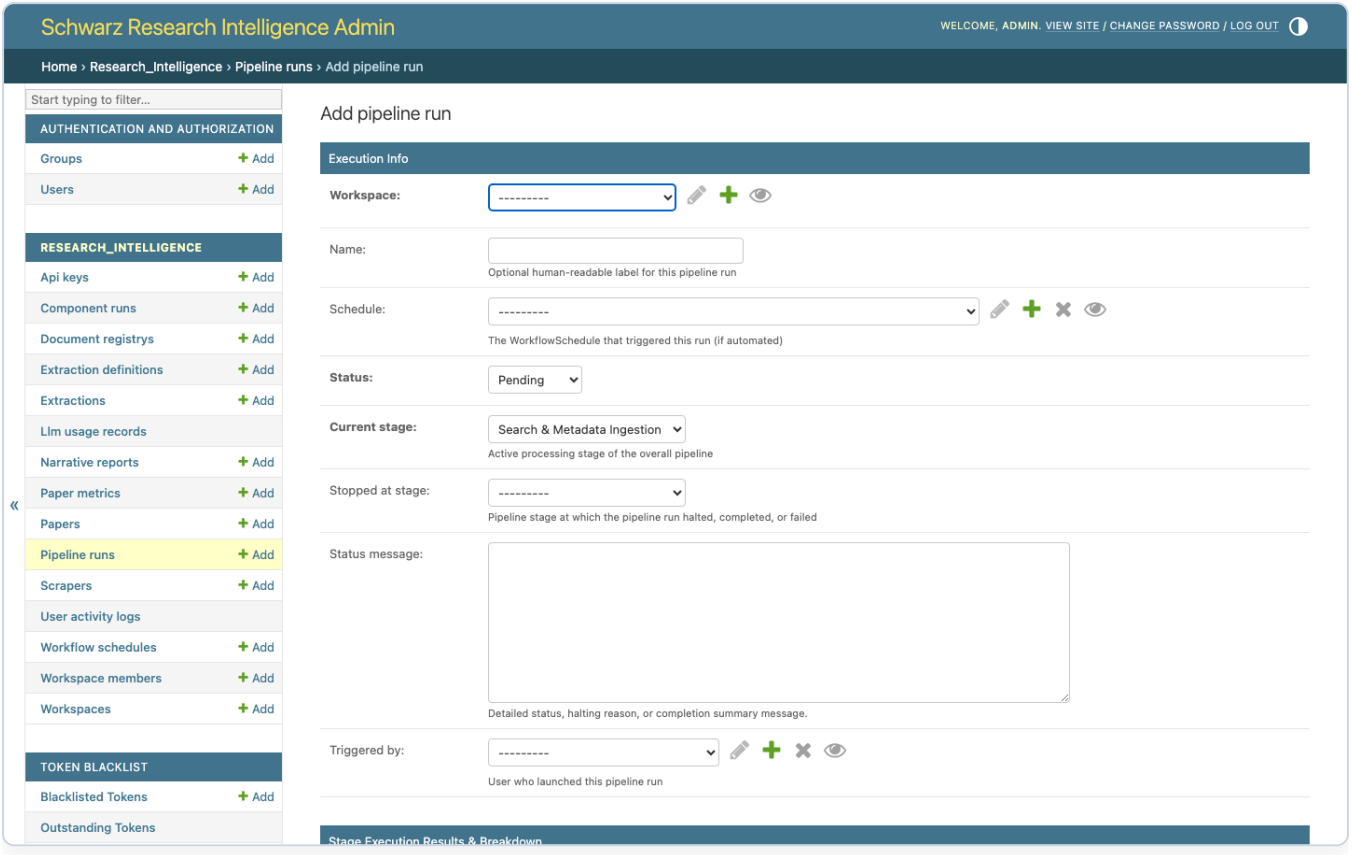


Figure 2: Pipeline Run creation modal with configuration override JSON and execution breakpoint controls.

4. Click **Save**.

The Celery distributed task dispatcher immediately picks up the run, sets `status="running"`, and launches the initial `search` ingestion worker.

🌐 Programmatic Launch via REST API

Managers can also trigger runs programmatically via REST API with date range parameters:

- **`POST /api/pipeline-runs/start/`**:

```
curl -X POST https://research-scanner.schwarz/api/pipeline-runs/start/ \
  -H "Authorization: Bearer <JWT_TOKEN>" \
  -H "Content-Type: application/json" \
  -d '{
    "workspace_id": "7b8e1a90-3f2d-4b5c-8a1e-9d0c2e3f4a5b",
    "name": "August 2026 Review",
    "start_date": "2026-08-01",
    "end_date": "2026-08-31",
    "n_last_days": 7
  }'
```

- `POST /api/workspaces/<workspace_id>/run-pipeline/`: Workspace-scoped endpoint accepting identical date parameters.

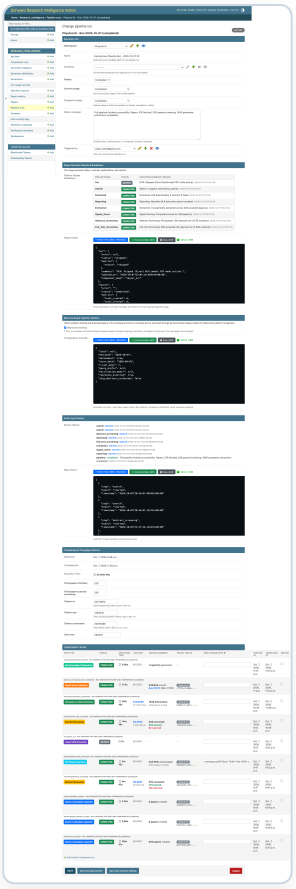


Figure 3: Active Pipeline Run detail page displaying live stage progress, stage results preview, step history preview, and linked Component Runs.

📊 Monitoring Execution in Real Time

You can monitor your research pipeline's progress directly from the Admin panel:

1. Key Metrics in the Pipeline Runs List

The consolidated changelist provides instant operational clarity:

Column	Description	Example
Papers (In ➔ Out)	Unified funnel metric displaying total harvested candidate papers vs. final papers that passed screening	`94 in → 27 out`
Status	Current high-level state of the pipeline run	`RUNNING`, `COMPLETED`, `FAILED`, `ABORTED`
Halt / Final Stage	Visual badge indicating completion status or the stage where execution paused	`✅ COMPLETED`, `⛔ ABSTRACT_SCREENING`, `⚠️ FAILED`
Status Message / Reason	Live progress narrative or error diagnostic explanation	`Full SLR pipeline finished successfully...`
Execution Time	Total run duration formatted for human readability	`4m 12s`, `43.0s`
USD Cost	Real-time aggregated Gemini API spending for this run	`$1.42`

2. Pipeline Run Detail View

Open any run record to inspect:

- **Stage Results Preview**: Formatted JSON summary of output counts per completed step (papers discovered, duplicates pruned, screening pass rates).
- **Step History Preview**: Chronological audit trail of all executed stages with individual duration measurements.
- **Component Runs Table**: Inline list of granular Celery worker tasks with links to raw storage paths and execution metrics.

3. Administrative Bulk Actions

From the Pipeline Runs changelist dropdown, administrators can execute batch operations on selected runs:

- 🔄 **Tick Selected Runs**: Advances the state machine for active or halted runs, triggering the next pending DAG stage.
- ▶️ **Resume Selected Runs**: Resumes runs that were stopped at an intermediate stage (e.g. after human review of abstract screening results).

🔄 Rerunning Pipeline from a Specific Step Forward

If you updated full-text criteria or adjusted an extraction schema, you don't need to re-scrape or re-download papers!

1. In the Admin sidebar, navigate to **Component Runs**.
2. Select the component run where you want execution to restart from (e.g. `Full-Text Screening` or `Extraction`).

3. Under the **Action** dropdown, select 🔄 **Rerun Pipeline from Selected Step Forward**.
4. Click **Go**.
5. The DAG engine resets downstream nodes, dispatches fresh Celery tasks, and continues execution automatically.

🎯 Next Steps

- Learn how to inspect technology scorecards and extracted parameter tables: **Guide 05: Viewing Results via Admin**.
- Learn how to monitor and trigger runs via Claude Code & MCP: **Guide 04: Running Pipelines via MCP**.

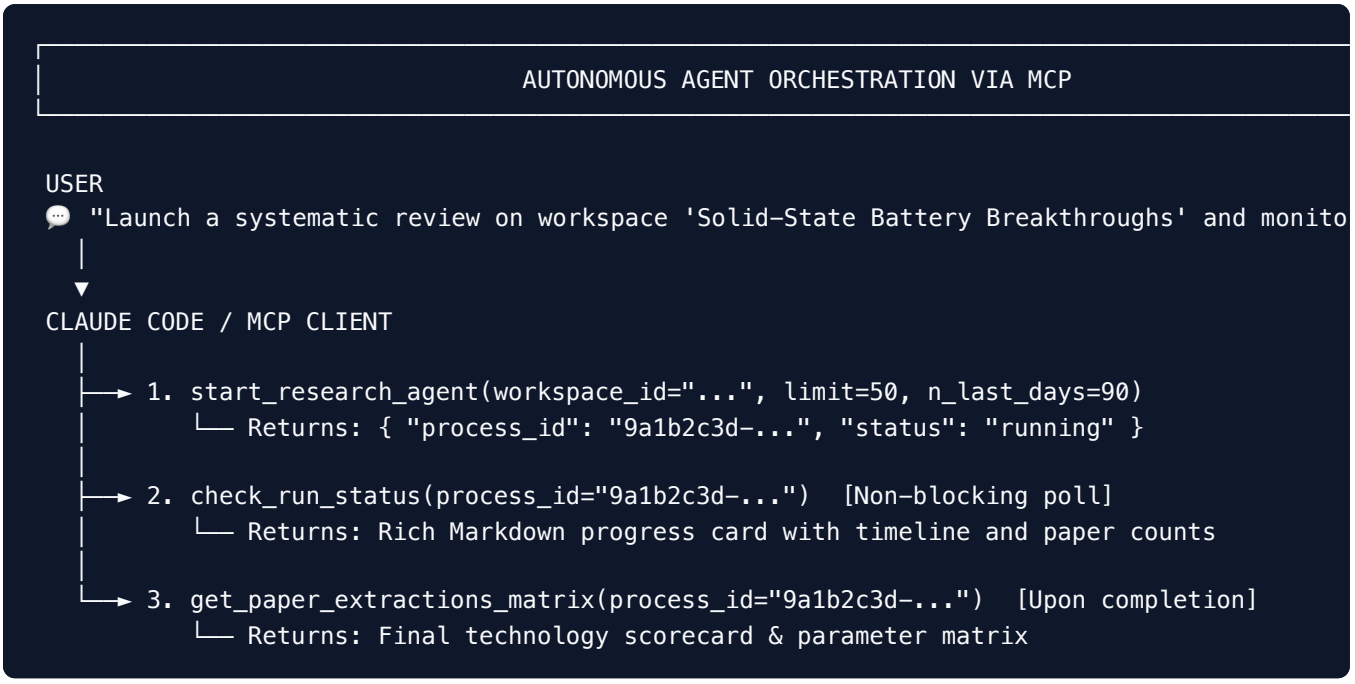
⚡ Guide 04: Running a Research Pipeline via MCP & Claude Code

With the **Schwarz Research Intelligence (SRI) Model Context Protocol (MCP)** server, you can trigger, monitor, pause, and supervise entire systematic reviews using Claude Code or any MCP-compatible AI agent.

🌐 Quick Access

- **Live MCP SSE Endpoint**: `https://research-scanner.schwarz/mcp/sse` *(or `https://sri-prod.apps.01.cf.eu01.onstackit.cloud/mcp/sse`)*
- **Web App URL**: https://research-scanner.schwarz/ *(or Cloud Foundry mirror)*
- **Admin Portal**: https://research-scanner.schwarz/admin/ *(or Cloud Foundry mirror)*

🔄 Autonomous Agent Orchestration Flow



💡 Quick Start: Conversational Execution

Simply ask Claude Code:

> **User Prompt**: *"Please start an autonomous research run for my workspace 'Solid-State Battery Breakthroughs 2026'. Limit ingestion to 50 papers from the last 90 days. Monitor the run and present the results when complete."*

Claude Code will launch the run and report real-time milestone updates.

🛠️ Detailed MCP Tool Reference

1. Launching an Autonomous Run (`start_research_agent`)

This is the **primary recommended entrypoint** for running systematic reviews.

Date Filtering Modes (Choose One):

You can filter harvested literature by publication date using **either** relative days **or** explicit calendar dates:

- **Mode 1: Relative Lookback Window (Default & Most Common)**
  - `n_last_days` *(optional, integer, default: 7)*: Number of past days to query (`now() - N days` to `now()`). Active when neither `start_date` nor `end_date` is specified.
  - *Example*: `"n_last_days": 14` (searches papers from the last 14 days).
- **Mode 2: Explicit Date Range**
  - `start_date` *(optional, ISO string, e.g. `'2026-08-01'`)*: Ingests papers published on or after this date.
  - `end_date` *(optional, ISO string, e.g. `'2026-08-31'`)*: Ingests papers published on or before this date. Defaults to `now()` if omitted when `start_date` is provided.
  - *Example*: `"start_date": "2026-08-01", "end_date": "2026-08-31"` (searches the exact August 2026 calendar interval).

General Parameters:

- `workspace_id` *(required)*: UUID of your target workspace.
- `limit` *(optional, default: null = all available)*: Maximum number of candidate papers to harvest across scrapers.
- `query_prefix` *(optional)*: Additional query keywords to prepend to all scraper queries.
- `skip_abstract_screening` *(optional, boolean)*: Set to `true` to skip abstract screening and route all papers directly to full text screening (default: inherits `workspace.skip_abstract_screening`).
- `reprocess_existing` *(optional, boolean)*: Reprocessing policy override. When omitted (`null`), inherits the workspace default policy (`workspace.reprocess_existing`). Set explicitly to `true` to re-screen and re-extract all existing workspace papers even if no new papers are found; set `false` to only process newly scraped papers.
- `notification_email` *(optional)*: Single-run email override to receive Mailgun completion report (default: sends to all active workspace Subscribers).

Example A: Relative Lookback Run (`start_research_agent` / `start_pipeline_run`):

```
{
  "workspace_id": "7b8e1a90-3f2d-4b5c-8a1e-9d0c2e3f4a5b",
  "name": "Weekly Autonomous Ingestion",
  "n_last_days": 14,
  "limit": 50
}
```

Example B: Explicit Calendar Interval Run (`start_research_agent` / `start_pipeline_run`):

```
{
  "workspace_id": "7b8e1a90-3f2d-4b5c-8a1e-9d0c2e3f4a5b",
  "name": "August 2026 Solid Electrolyte Review",
  "start_date": "2026-08-01",
  "end_date": "2026-08-31",
  "limit": 50,
  "reprocess_existing": false,
  "notification_email": "innovation-team@schwarz.com"
}
```

Response:

```
{
  "status": "success",
  "process_id": "9a1b2c3d-4e5f-6a7b-8c9d-0e1f2a3b4c5d",
  "pipeline_run_id": "e4f5a6b7-c8d9-0e1f-2a3b-4c5d6e7f8a9b",
  "message": "Autonomous Research Agent Process started successfully."
}
```

Granular Search Trigger (`trigger_search_job`):

To execute Stage 1 Search as a standalone step:

```
{
  "workspace_id": "7b8e1a90-3f2d-4b5c-8a1e-9d0c2e3f4a5b",
  "start_date": "2026-08-01",
  "end_date": "2026-08-15",
  "limit": 25
}
```

(Or with relative days):

```
{
  "workspace_id": "7b8e1a90-3f2d-4b5c-8a1e-9d0c2e3f4a5b",
  "n_last_days": 30,
  "limit": 25
}
```

2. Monitoring Real-Time Status (`check_run_status`)

`check_run_status` is the **primary non-blocking status inspector**. It returns live metrics, percentage completion, fine-grained TQDM progress bars, and a rendered ASCII progress card.

Tool Invocation Example:

```
{
  "process_id": "9a1b2c3d-4e5f-6a7b-8c9d-0e1f2a3b4c5d"
}
```

Example Output Markdown Card:

```
### 🚀 SLR Pipeline Status: `running` (65%)
**Process ID**: `9a1b2c3d-4e5f-6a7b-8c9d-0e1f2a3b4c5d`
**Elapsed Time**: `1m 42s` | **Cost**: `$0.14 USD` | **Tokens**: `142,500`

`[████████████████████▒▒▒▒▒▒▒▒▒▒] 65% Complete`

#### 📋 Execution Stages:
| Stage | Status | Details |
| :--- | :--- | :--- |
| **1. Search & Ingestion** | ✅ Done (4.2s) | 50 papers discovered (ArXiv: 25, OpenAlex: 2
| **2. Abstract Screening** | ✅ Done (18.1s)| 42 screened (18 passed, 24 rejected) |
| **3. Parallel PDF Download**| ✅ Done (12.4s)| 18 PDFs stored (0 paywalled, SHA-256 cache
| **4. Layout OCR Parsing** | ⏭️ Skipped | Direct multimodal Gemini 3.7 Flash active |
| **5. Full-Text Screening** | 🔄 In Progress | [██████▒▒▒▒▒▒] (12/18 items • 66.7%) |
| **6. Dynamic Extraction** | ⏳ Pending | Awaiting full-text screening completion |
| **7. Signal Score Ranking**| ⏳ Pending | Python scoring formula queued |
| **8. Executive Reporting** | ⏳ Pending | Report compilation & email digest queued |
```

3. State Management Tools

Tool Name	Parameters	Description
`pause_research_agent`	`process_id`	Pauses the active process without losing state or downloaded files.
`resume_research_agent`	`process_id`	Resumes a paused process from the exact active step.
`abort_research_agent`	`process_id`	Cancels the active process and terminates background Celery workers.
`list_research_agents`	`workspace_id`	Lists all historical and active processes with runtimes and completion states.
`get_research_agent_status`	`process_id`	Returns low-level diagnostic logs, task IDs, and storage footprints.
`get_pipeline_logs`	`process_id`, `limit`	Live terminal telemetry and worker logs inspector.

🎯 Next Steps

When `check_run_status` reports `is_finished: true` and `status: completed`:

- View the aggregated parameter scorecard: **Guide 06: Viewing Results via MCP**
- Chat with accepted papers: **Guide 07: Chatting with Papers via MCP**

📊 Guide 05: Viewing Review Results & Scorecards via Django Admin

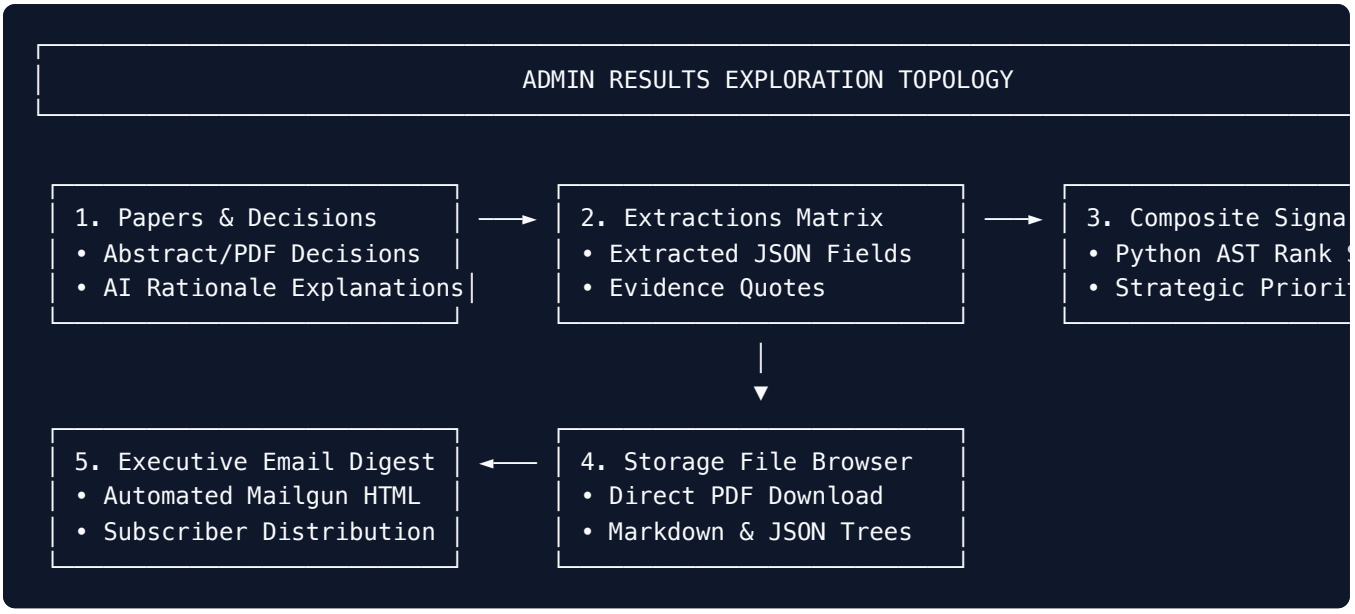
This guide details how to inspect, analyze, and audit Systematic Literature Review (SLR) results, technology scorecards, and parameter extractions using the **Schwarz Research Intelligence (SRI) Admin Portal**.

🌐 Quick Access

- **Admin Portal URL**: https://research-scanner.schwarz/admin/ *(or Cloud Foundry mirror)*
- **Storage Browser**: https://research-scanner.schwarz/admin/storage-browser/ *(or Cloud Foundry mirror)*

🧭 Navigating Results in Django Admin

When an autonomous review finishes, rich intelligence data is available across four main sections:



📄 1. Inspecting Screened Papers & AI Decision Rationales



Figure 1: Screened Papers list showing bibliographic metadata, DOI, status, and signal scores.

1. In the Admin sidebar, navigate to **Research Intelligence ➡️ Papers**.
2. Filter papers by:
   - **Workspace**: Select your workspace.
   - **Screening Status**: `accepted` (passed both abstract and full-text screening) or `rejected`.
   - **Pipeline Run**: Filter by specific review run.

Paper Detail View:

Click on any paper title to view its complete audit card:

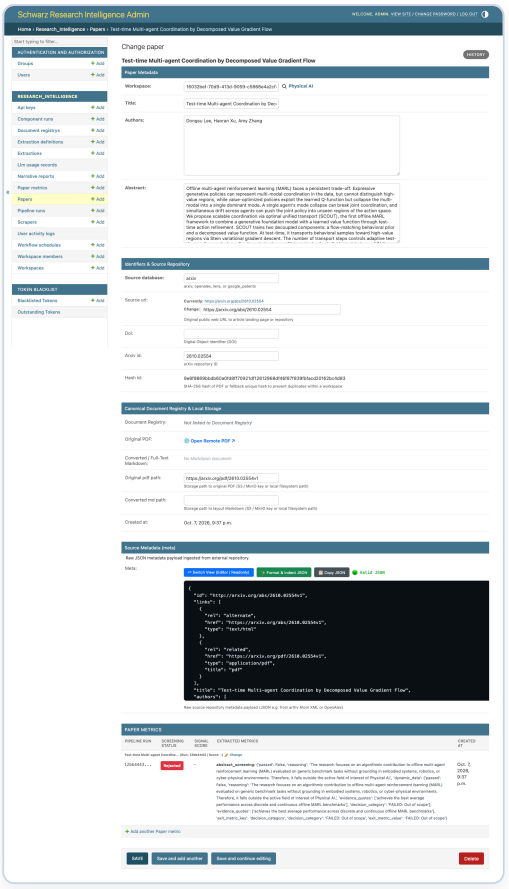


Figure 2: Paper detail page displaying abstract, screening status, decision category, and Gemini 3.7 Flash decision reasoning.

- **Bibliographic Metadata**: Title, authors, DOI, ArXiv ID, abstract, and publication date.
- **Screening Status**: `accepted` | `rejected` | `pending` | `failed`.
- **Screening Decision Category**: e.g., `include_full_text`, `exclude_hardware_only`, `exclude_survey`.
- **AI Decision Rationale**: Transparent explanation of why Gemini 3.7 Flash accepted or rejected the document against your exact criteria:

> *"Accepted: The paper presents a novel QAOA formulation for multi-depot vehicle routing with 48 qubits, reporting a 14.2% circuit depth reduction compared to standard QAOA baselines (Section 4, Page 6)."*

🔍 2. Reviewing Extracted Parameters & Evidence Quotes

To examine deep technical KPIs and parameters extracted during Step 5:



Figure 3: Extractions matrix showing extracted JSON schemas, confidence scores, and validation flags.

1. Click **Extractions** in the Admin sidebar.
2. Filter by **Definition** (e.g. `Battery Specs`, `Schwarz Strategic Fit`, `TRL Assessment`).

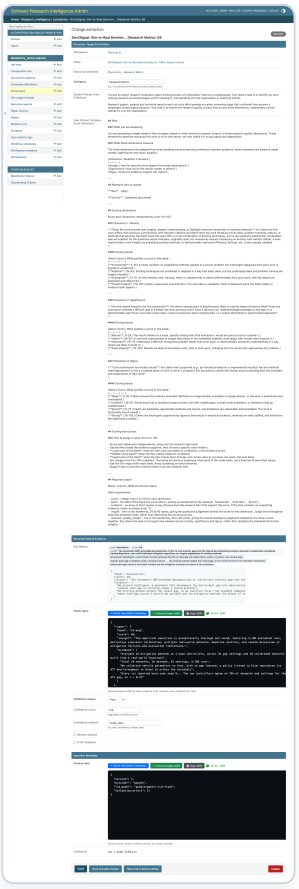


Figure 4: Detailed view of an extraction record with raw LLM output, JSON values, and verbatim evidence quote.

3. Each extraction record contains:
   - **Paper**: Linked academic paper.
   - **Extracted Data (JSON)**:

```
{
  "electrolyte_type": "Sulfide",
  "energy_density_wh_kg": 432.5,
  "room_temp_conductivity_ms_cm": 12.8,
  "cycle_retention_pct": 88.4,
  "cycles_tested": 500,
  "evidence_quote": "The fabricated pouch cell demonstrated a gravimetric energy dens
}
```

   - **Confidence Score**: Model confidence rating (e.g. `0.96`).
   - **Validation Status**:
     - `pass` — Schema strictly validated and evidence verified.
     - `warn` — Minor field coercion or edge value.
     - `fail` — Missing mandatory schema constraints.

🧱 3. Reviewing Composite Signal Scores & Paper Rankings

Papers are automatically ranked according to your workspace's custom Python AST mathematical formula (`signal_score_code`).

1. Navigate to **Paper Metrics** or sort the **Papers** list by `signal_score` (Descending).
2. Example Scorecard Table:

Rank	Paper Title	Authors	Fit Score	TRL	Key Metric	Signal Score
🥇 1	*High-Conductivity Sulfide Solid Electrolytes for 450 Wh/kg Pouch Cells*	Zhang et al. (2026)	9.5 / 10	5	432.5 Wh/kg	**94.2 / 100**
🥈 2	*Halide-Garnet Composite Interlayers for Dendrite-Free Anodes*	Müller et al. (2026)	8.8 / 10	4	395.0 Wh/kg	**82.5 / 100**
🥉 3	*Scalable Dry-Coating Process for Polymer Solid-State Batteries*	Chen et al. (2025)	8.0 / 10	6	340.0 Wh/kg	**76.8 / 100**

Click into **Paper Metric** to see the calculation variable audit trail.

📦 4. Using the Admin Storage Browser

The built-in **Storage Browser** allows administrators and researchers to inspect and download all raw artifacts stored on disk/S3.

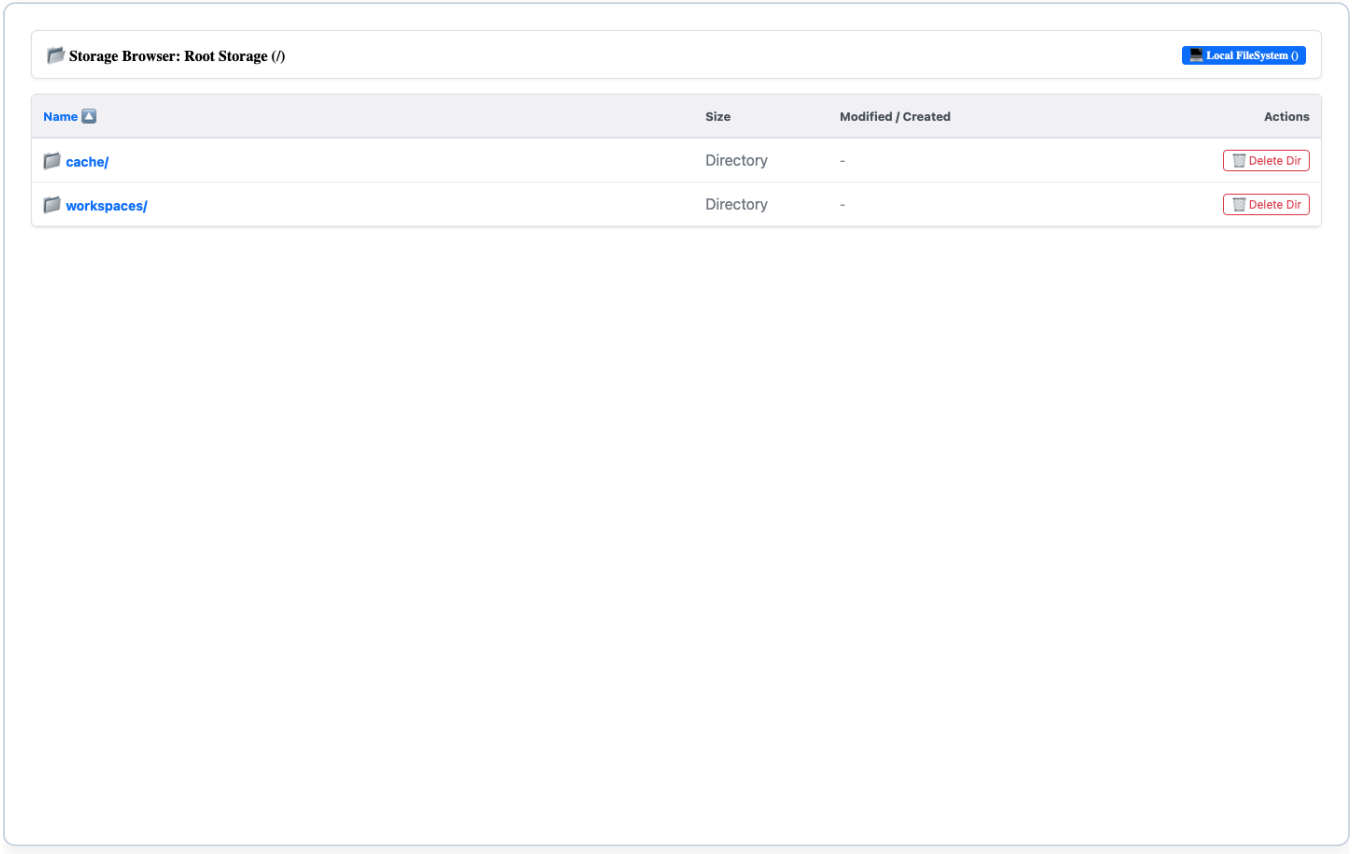


Figure 5: Django Storage Browser allowing direct download of original PDFs, layout Markdown, and JSON metadata.

1. Open https://sri-prod.apps.01.cf.eu01.onstackit.cloud/admin/storage-browser/.
2. Navigate the workspace directory tree:

```
workspaces/
└── 7b8e1a90-3f2d-4b5c-8a1e-9d0c2e3f4a5b/
    └── runs/
        └── e4f5a6b7-c8d9-0e1f-2a3b-4c5d6e7f8a9b/
            ├── documents/
            │   ├── 8f14a9c2.../
            │   │   ├── original.pdf          [Download PDF]
            │   │   ├── ocr_layout.md         [View Markdown]
            │   │   └── extractions.json      [View Raw JSON]
            └── reports/
                └── executive_briefing.pdf    [Download Report]
```

3. Click any `.pdf`, `.md`, or `.json` file to download or preview directly in your browser.

✉️ 5. Dispatching Executive Email Digests

When a run is completed, an executive summary email can be sent to all workspace subscribers:

1. In the Admin panel, navigate to **Pipeline Runs**.
2. Select your completed run.
3. In the **Action** dropdown at the top, select **Send Executive Email Digest to Subscribers**.
4. Click **Go**.
5. SRI automatically renders a responsive HTML email via Mailgun containing KPI counters, executive narrative briefings, and top-ranked papers.

🎯 Next Steps

- To view and export scorecard tables in Claude Code: **Guide 06: Viewing Results via MCP**
- To deep-dive and interrogate any paper with AI: **Guide 07: Chatting with Papers via MCP**

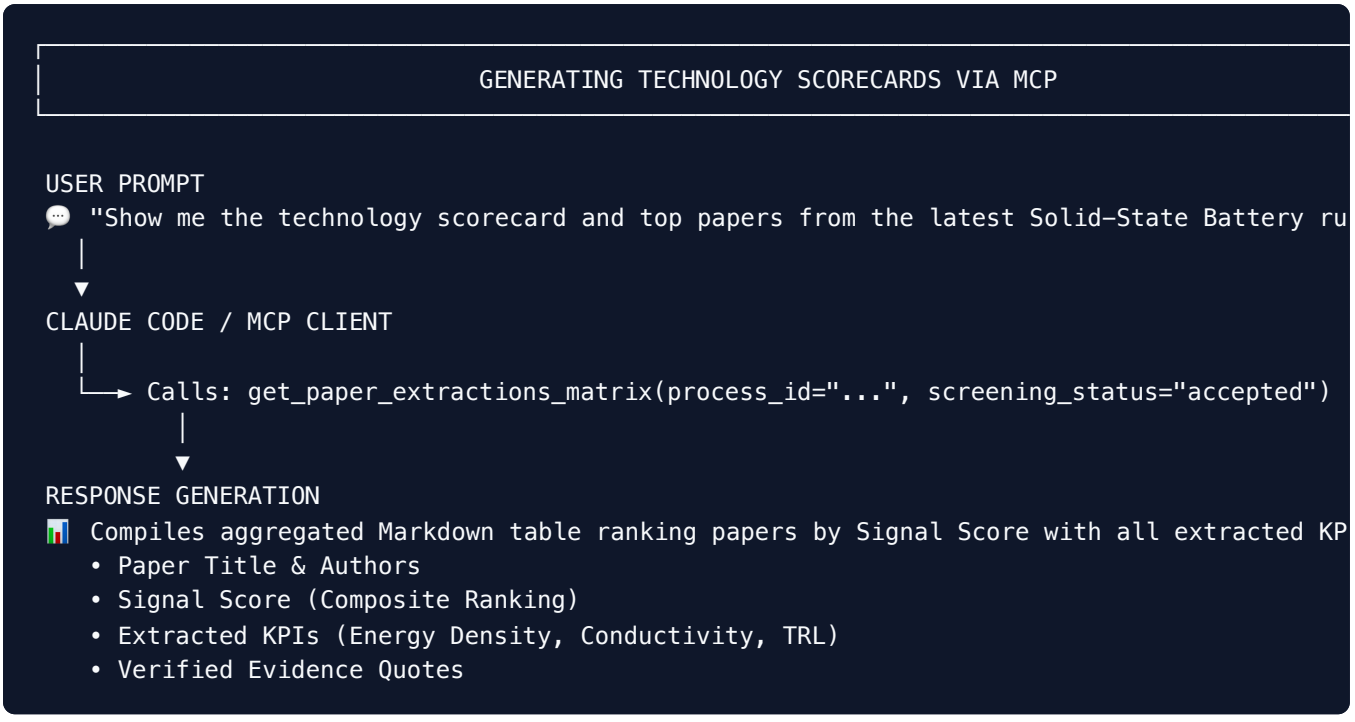
📊 Guide 06: Viewing Review Results & Technology Scorecards via MCP

Using Claude Code, Cursor, or any MCP AI assistant, you can instantly generate structured technology scorecards, compare scientific performance metrics, and inspect accepted literature in formatted Markdown tables.

🌐 Quick Access

- **Live MCP SSE Endpoint**: `https://research-scanner.schwarz/mcp/sse` *(or `https://sri-prod.apps.01.cf.eu01.onstackit.cloud/mcp/sse`)*
- **Web App URL**: https://research-scanner.schwarz/ *(or Cloud Foundry mirror)*
- **Admin Portal**: https://research-scanner.schwarz/admin/ *(or Cloud Foundry mirror)*

🧭 Scorecard Query Flow



💡 Quick Start: Conversational Scorecard Prompts

You can request results in natural language:

Prompt 1: High-Level Technology Scorecard

> *"Show me the technology matrix and extracted parameters for all accepted papers in workspace 'Solid-State Battery Breakthroughs 2026'."*

Prompt 2: Parameter Comparison Table

> *"Compare the ionic conductivity, energy density, and TRL across all accepted papers. Highlight the top 3 highest-scoring technologies."*

Prompt 3: Deep Screening Audit

> *"List the papers that were rejected during full-text screening along with Gemini's rationale for rejection."*

🛠️ Primary MCP Tools for Results & Extractions

1. `get_paper_extractions_matrix` (Primary Scorecard Tool)

This is the **core tool** for retrieving run results. It aggregates all accepted papers, their composite signal scores, screening decisions, and dynamic extraction definitions in a **single structured call**.

Parameters:

- `process_id` *(optional)*: Filter by Research Agent Process UUID.
- `pipeline_run_id` *(optional)*: Filter by Pipeline Run UUID.
- `workspace_id` *(optional)*: Filter by Workspace UUID.
- `screening_status` *(optional)*: Filter by `accepted` (default) or `rejected`.
- `limit` *(optional)*: Max number of papers to return.

Example Tool Invocation:

```
{
  "workspace_id": "7b8e1a90-3f2d-4b5c-8a1e-9d0c2e3f4a5b",
  "screening_status": "accepted"
}
```

Example Output Rendered by Claude Code:

🏆 Technology Assessment Scorecard

Rank	Paper Title	Authors	Energy Density	Conductivity	TRL	Signal Score
🥇 1	**High-Conductivity Sulfide Solid Electrolytes for 450 Wh/kg Pouch Cells**	Zhang, L., et al. (2026)	**432.5 Wh/kg**	12.8 mS/cm	**5**	**94.2**
🥈 2	**Halide-Garnet Composite Interlayers for Dendrite-Free Anodes**	Müller, H., et al. (2026)	**395.0 Wh/kg**	8.4 mS/cm	**4**	**82.5**
🥉 3	**Scalable Dry-Coating Process for Polymer Solid-State Batteries**	Chen, W., et al. (2025)	**340.0 Wh/kg**	3.2 mS/cm	**6**	**76.8**

🔍 **Detailed Findings & Evidence Quotes:**

1. **High-Conductivity Sulfide Solid Electrolytes for 450 Wh/kg Pouch Cells**
   - **DOI / ArXiv**: `arXiv:2602.04182` | **Status**: `accepted`
   - **Signal Score**: `94.2 / 100`
   - **Battery Specs**:
     - *Electrolyte Type*: Sulfide ($Li_{10}GeP_2S_{12}$ derivative)
     - *Energy Density*: 432.5 Wh/kg
     - *Room Temp Conductivity*: 12.8 mS/cm
     - *Cycle Retention*: $88.4%\text{ after } 500\text{ cycles}$
   - **Verbatim Evidence Quote**:

     > *"The fabricated pouch cell demonstrated a gravimetric energy density of 432.5 Wh kg−1 at 0.1 C and retained 88.4% capacity after 500 cycles at room temperature." (Page 4, Table 2)*

   - **TRL Assessment**: Level 5 (Component validation in relevant environment / pouch cell).

---

2. `list_papers` (Bibliographic Explorer)

List all ingested papers with metadata, abstracts, and screening statuses:

```
{
  "workspace_id": "7b8e1a90-3f2d-4b5c-8a1e-9d0c2e3f4a5b",
  "limit": 20
}
```

3. `list_extractions` & `get_extraction` (Low-Level Audit)

Inspect raw extraction objects, confidence scores, and validation flags:

```
{
  "workspace_id": "7b8e1a90-3f2d-4b5c-8a1e-9d0c2e3f4a5b",
  "validation_status": "pass",
  "limit": 10
}
```

🎯 Next Steps

Want to interrogate a specific paper on its experimental methodology, verify circuit depths, or ask follow-up questions?

- Deep dive with AI chat: **Guide 07: Chatting with Papers via MCP**

💬 Guide 07: Interrogating & Chatting with Papers via MCP (Ask Paper & Knowledge Explorer)

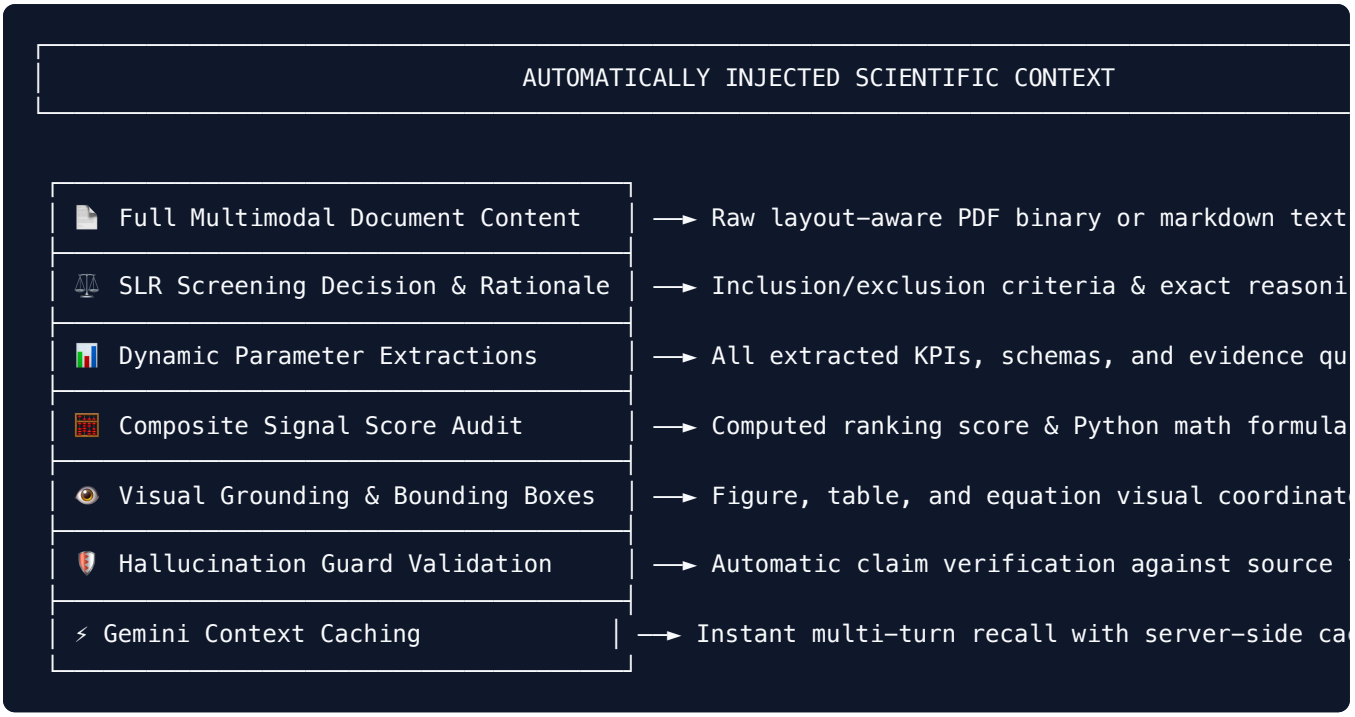
The **Schwarz Research Intelligence (SRI)** interactive paper exploration tool (`ask_paper`) turns ingested academic papers into dynamic, conversational scientific advisors powered by **Google Gemini 3.7 Flash**.

🌐 Quick Access

- **Live MCP SSE Endpoint**: `https://research-scanner.schwarz/mcp/sse` *(or `https://sri-prod.apps.01.cf.eu01.onstackit.cloud/mcp/sse`)*
- **Web App URL**: https://research-scanner.schwarz/ *(or Cloud Foundry mirror)*
- **Admin Portal**: https://research-scanner.schwarz/admin/ *(or Cloud Foundry mirror)*

🧠 What Makes SRI Paper Chat Unique?

Unlike standard document chat tools that only pass text fragments, `ask_paper` injects the **complete scientific context** of the review:



🧭 Multi-Turn Conversation Workflow

```
Turn 1 (Initiate Chat Session)
┌──────────────────────────────────────────────────────────────────────────────
│ USER: "Explain how the authors fabricated the solid electrolyte pouch cell and verify co
│ └─► ask_paper(workspace_id=..., paper_id=..., question=..., deep_thinking=True)
│     └─► Returns Answer + Citations + session_id: "c8e1a90b-..."
└──────────────────────────────────────────────────────────────────────────────
                                      │
                                      ▼
Turn 2+ (Continue Conversational Context)
┌──────────────────────────────────────────────────────────────────────────────
│ USER: "Does this match the impedance spectroscopy shown in Figure 3?"
│ └─► ask_paper(workspace_id=..., session_id="c8e1a90b-...", question=...)
│     └─► Reuses server-cached tokens for instant, zero-cost conversational memory
└──────────────────────────────────────────────────────────────────────────────
```

💡 Real-World Prompts to Ask Claude Code

1. Methodology & Experimental Verification

> *"In Zhang et al. (2026), what exact solvent and sintering temperature did they use for the sulfide electrolyte membrane? Point to the exact table or figure."*

2. Deep Mathematical & Thermodynamic Derivations (`deep_thinking=True`)

> *"Derive the mathematical proof used in Section 3 for the quantum circuit upper bound. Verify if the asymptotic complexity claim $O(N \log N)$ holds under their error model."*

3. Cross-Paper Synthesis (Workspace Overview)

> *(Omit `paper_id`) "Across all accepted papers in this workspace, summarize the main degradation mechanisms identified for lithium metal anodes under fast charging (>2C)."*

4. SLR Decision Deep-Dive

> *"Why did the SLR pipeline accept paper A but reject paper B? Compare their reported volumetric energy densities against our workspace criteria."*

🛠️ MCP Tool: `ask_paper` Reference

Parameters:

- `workspace_id` *(required)*: UUID of the workspace.
- `question` *(required)*: Detailed research question or instruction.
- `paper_id` *(optional)*: Target Paper UUID. *Omit for a workspace overview: the model then sees only the titles and signal scores of the top accepted papers, not their text.*
- `session_id` *(optional)*: UUID of an existing conversation session to continue.
- `deep_thinking` *(optional, boolean)*: Set to `true` to raise the model's thinking budget, get its thought trace back in `thinking_process`, and have it verify numerical claims against the document. Slower and costs more output tokens.

Tool Invocation Example (Turn 1):

```
{
  "workspace_id": "7b8e1a90-3f2d-4b5c-8a1e-9d0c2e3f4a5b",
  "paper_id": "3a4b5c6d-7e8f-9a0b-1c2d-3e4f5a6b7c8d",
  "question": "What is the primary degradation mechanism of the sulfide electrolyte at 4.5V
  "deep_thinking": true
}
```

Tool Response Payload:

```
{
  "status": "completed",
  "session_id": "c8e1a90b-4f2d-4b5c-8a1e-9d0c2e3f4a5b",
  "message_id": "1f2e3d4c-5b6a-7890-bcde-f0123456789a",
  "paper_id": "3a4b5c6d-7e8f-9a0b-1c2d-3e4f5a6b7c8d",
  "paper_title": "High-Conductivity Sulfide Solid Electrolytes for 450 Wh/kg Pouch Cells",
  "answer": "Above 4.2 V the sulfide framework oxidises [Fig 4, quote: \"irreversible oxida
  "thinking_process": "...",
  "citations": [],
  "visual_citations": [
    { "label": "Fig 4", "quote": "irreversible oxidation peaks beginning at 2.3 V with seve
    { "label": "Table 2", "quote": "88.4% capacity retention after 500 cycles", "verified_i
  ],
  "hallucination_guard": { "verified": true, "grounding_score": 1.0, "verified_count": 2, "
  "deep_thinking": true,
  "llm_model": "google/gemini-3.8-flash",
  "tokens_in": 18421,
  "tokens_out": 1930,
  "tokens_cached": 17800,
  "total_tokens": 20351,
  "usd_cost": 0.0112,
  "session_total_tokens": 20351,
  "session_total_usd_cost": 0.0112,
  "workspace_monthly_tokens": 1483902,
  "workspace_monthly_usd": 4.87
}
```

Markers carry a page number (`[Page 5, Fig 4, quote: "..."]`) and a `box_2d` only when the paper was sent as a PDF; the usual OCR-markdown path has no page numbers. Generation failures return `{ "status": "failed", "message_id", "session_id", "error" }`; bad requests return `{ "status": "error", "error" }`.

🎯 Summary

With `ask_paper` and the SRI platform, enterprise research teams can:

1. Extract precise engineering parameters in minutes.
2. Verify claims directly against original PDF figures and equations.
3. Rapidly synthesize evidence across hundreds of academic publications.

For full system details and pipeline configuration, explore the complete User Documentation Index.